MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL,

TTL, DATA SELECTORS/MULTIPLEXERS,

MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic, silicon, TTL, data selectors/multiplexers, logic microcircuits. Three product assurance classes and a choice of case outlines and lead finishes are provided for each type and are reflected in the complete part number.

1.2 Part number. The complete part number shall be in accordance with MIL-M-38510.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	Sixteen-input data selector/multiplexer, with enable
02, 06	Eight-input data selector/multiplexer, with enable
03	Dual, four-input data selector/multiplexer, with enable
04	Dual, four-input data selector/multiplexer, without enable
05	Quad, two-input data selector/multiplexer, with enable

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Outline letter	Case outline, (see MIL-M-38510, appendix C)
E	D-2 (16-lead, 1/4" x 7/8", dual-in-line package)
F	F-5 (16-lead, 1/4" x 3/8", flat package)
J	D-3 (24-lead, 1/2" x 1-1/4", dual-in-line package)
K	F-6 (24-lead, 3/8" x 5/8", flat-package)
Z	F-8 (24-lead, 1/4" x 3/8", flat-package)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center, RBE-2, Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.3 Absolute maximum ratings.

Supply voltage range	-0.5 V to 7.0 V
Input voltage range	-1.5 V at -12 mA to 5.5 V
Storage temperature range	-65°C to 150°C
Maximum power dissipation, P_D 1/	
Device type 01	375 mW
Device types 02 and 06	268 mW
Device type 03	286 mW
Device type 04	248 mW
Device type 05	275 mW
Lead temperature (soldering 10 seconds)	300°C
Thermal resistance, junction to case (θ_{JC})	0.09°C/mW cases F, K, and Z 0.08°C/mW cases E and J
Junction temperature (T_J)	175°C

1.4 Recommended operating conditions.

Supply voltage	4.5 V minimum to 5.5 V maximum
Minimum high level input voltage (V_{IH})	2.0 V
Maximum low level input voltage (V_{IL})	0.8 V
Maximum low level output current (I_{OL})	16 mA
Normalized fanout (each output) 2/	
Low logic level	10 maximum
High logic level	20 maximum
Case operating temperature range	-55°C to 125°C

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuit, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order or precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, contruction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-39510, and herein.

1/ Must withstand the added P_D due to short circuit condition (e.g. I_{OS} test.)

2/ Device will fanout in both high and low levels to the specified number of inputs on the same device type as that being tested.

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3.2.1 Logic diagrams and terminal connections. The logic diagrams and terminal connections shall be as specified on figures 1 and 2.

3.2.2 Truth tables. The truth tables shall be as specified on figure 3.

3.2.3 Schematic circuit. The schematic circuit shall be submitted to the preparing activity prior to inclusion of manufacturer's device in this specification and shall be submitted to the qualifying activity as a prerequisite for qualification. All manufacturer's schematics shall be maintained and available upon request.

3.2.4 Case outlines. The case outlines shall be as specified in 1.2.3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.5).

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range, unless otherwise specified.

TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions	Device type	Limits		Units
				Min	Max	
High level output voltage	V_{OH}	V_{CC} = 4.5 V I_{OH} = -.8 mA	01, 02, 03, 04, 05, 06	2.4		Volts
Low level output voltage	V_{OL}	V_{CC} = 4.5 V I_{OL} = 16 mA	01, 02, 03, 04, 05, 06		0.4	Volts
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V I_{IN} = -12 mA	01, 02, 03, 04, 05, 06		-1.5	Volts
Low level input current	I_{IL}	V_{CC} = 5.5 V V_{IN} = 0.4 V	01, 02, 03, 04, 05, 06	-0.7	-1.6	mA
High level input current	I_{IH1}	V_{CC} = 5.5 V V_{IN} = 2.4 V	01, 02, 03, 04, 05, 06		40	µA
High level input current	I_{IH2}	V_{CC} = V_{IN} = 5.5 V	01, 02, 03, 04, 05, 06		100	µA
Short circuit output current	I_{OS}	V_{CC} = 5.5 V V_{OUT} = 0 V 1/	01, 03, 06	-20	-55	mA
			02, 04, 05	-20	-120	mA
Supply current	I_{CC}	V_{CC} = 5.5 V	01		68	mA
			02, 06		48	mA
			04		45	mA
			03		52	mA
			05		50	mA
Propagation delay time high-to-low level output from A, B, C or D to W	t_{PHL1}	R_L = 390Ω ±5%, C_L = 50 pF minimum (figure 5)	01	8	40	ns
Propagation delay time low-to-high level output from A, B, C or D to W	t_{PLH1}		01	8	43	ns
Propagation delay time high-to-low level output from strobe to W	t_{PHL2}		01	6	37	ns
Propagation delay time low-to-high level output from strobe to W	t_{PLH2}		01	6	32	ns
Propagation delay time high-to-low level output from E_0-E_{15} to W	t_{PHL3}		01	3	23	ns
Propagation delay time low-to-high level output from E_0-E_{15} to W	t_{PLH3}		01	3	30	ns

1/ Not more than one should be shorted at one time.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions	Device type	Limits		Units
				Min	Max	
Propagation delay time high-to-low level output from A, B or C to W	t_{PHL1}	R_L = 390Ω ±5%, C_L = 50 pF minimum (figure 8)	02, 06	6	40	ns
Propagation delay time low-to-high level output from A, B or C to W	t_{PLH1}		02, 06	6	38	ns
Propagation delay time high-to-low level output from A, B or C to Y	t_{PHL2}		02, 06	8	49	ns
Propagation delay time low-to-high level output from A, B or C to Y	t_{PLH2}		02, 06	8	45	ns
Propagation delay time high-to-low level output from strobe to W	t_{PHL3}		02, 06	6	35	ns
Propagation delay time low-to-high level output from strobe to W	t_{PLH3}		02, 06	6	35	ns
Propagation delay time high-to-low level output from strobe to Y	t_{PHL4}		02, 06	8	46	ns
Propagation delay time low-to-high level output from strobe to Y	t_{PLH4}		02, 06	8	42	ns
Propagation delay time high-to-low level output from D_0-D_7 to W	t_{PHL5}		02, 06	3	32	ns
Propagation delay time low-to-high level output from D_0-D_7 to W	t_{PLH5}		02, 06	3	26	ns
Propagation delay time high-to-low level output from D_0-D_7 to Y	t_{PHL6}		02, 06	6	41	ns
Propagation delay time low-to-high level output from D_0-D_7 to Y	t_{PLH6}		02, 06	6	33	ns
Propagation delay time high-to-low level output from data to Y	t_{PHL1}	R_L = 390Ω ±5%, C_L = 50 pF minimum (figure 7)	03	3	29	ns
Propagation delay time low-to-high level output from data to Y	t_{PLH1}		03	3	28	ns
Propagation delay time high-to-low level output from A or B to Y	t_{PHL2}		03	6	44	ns
Propagation delay time low-to-high level output from A or B to Y	t_{PLH2}		03	6	42	ns
Propagation delay time high-to-low level output from strobe to Y	t_{PHL3}		03	6	32	ns
Propagation delay time low-to-high level output from strobe to Y	t_{PLH3}		03	6	42	ns

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions	Device type	Limits		
				Min	Max	Units
Propagation delay time high-to-low level output from data to Y	t_{PHL1}	R_L = 390Ω ±5%, C_L = 50 pF minimum (figure 7)	04	3	41	ns
Propagation delay time low-to-high level output from data to Y	t_{PLH1}		04	3	39	ns
Propagation delay time high-to-low level output from data to W	t_{PHL2}		04	3	25	ns
Propagation delay time low-to-high level output from data to W	t_{PLH2}		04	3	24	ns
Propagation delay time high-to-low level output from A or B to Y	t_{PHL3}		04	6	51	ns
Propagation delay time low-to-high level output from A or B to Y	t_{PLH3}		04	6	51	ns
Propagation delay time high-to-low level output from A or B to W	t_{PHL4}		04	6	39	ns
Propagation delay time low-to-high level output from A or B to W	t_{PLH4}		04	6	34	ns
Propagation delay time high-to-low level output from A to Y	t_{PHL1}	R_L = 390Ω ±5%, C_L = 50 pF minimum (figure 8)	05	6	49	ns
Propagation delay time low-to-high level output from A to Y	t_{PLH1}		05	6	41	ns
Propagation delay time high-to-low level output from strobe to Y	t_{PHL2}		05	3	39	ns
Propagation delay time low-to-high level output from strobe to Y	t_{PLH2}		05	3	33	ns
Propagation delay time high-to-low level output from data to Y	t_{PHL3}		05	3	25	ns
Propagation delay time low-to-high level output from data to Y	t_{PLH3}		05	3	35	ns

3.5 Electrical test requirements. The electrical test requirements for each device class shall be in the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510 and 1.2 herein. At the option of the manufacturer, marking of the country of origin may be omitted from the body of the microcircuit, but shall be retained on the initial container.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 4 (see MIL-M-38510, appendix E).

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TABLE II. Electrical test requirements.

MIL-STD-883 Test requirements	Subgroups (see table III)		
	Class S devices	Class B devices	Class C devices
Interim electircal tests (pre burn-in) (method 5004)	1	1	None
Final electrical tests (method 5004)	1*,2,3,7, 9,10,11,	1*,2,3,7, 9	1,7
Group A electrical tests (method 5005)	1,2,3,7,8, 9,10,11	1,2,3,7, 9,10,11	1,2,3,7, 9
Group B test requirements (method 5005) subgroup 5	1,2,3,7,8, 9,10,11	N/A	N/A
Group C end-point electrical tests (method 5005)	N/A	1,2,3	1
Additional electrical tests for group C inspection	N/A	None	10,11
Group D end-point electrical tests (method 5005)	1,2,3	1,2,3	1

*PDA applies to subgroup 1 (see 4.2c).

4. PRODUCT ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspections. The following additional criteria shall apply:

a. Burn-in test (method 1015 of MIL-STD-883).
   (1) Test condition D or E, using the circuit shown on figure 5, or equivalent.
   (2) T_A = 125°C minimum.
b. Interim and final electrical tests shall be as specified in table II, except interim electrical test prior to burn-in is optional at the discretion of the manufacturer.
c. The percent defective allowable (PDA) for class S devices shall be as specified in MIL-M-38510. The PDA for class B devices shall be 10 percent based on failures from group A, subgroup 1 tests after cooldown as the final electrical test in accordance with method 5004 of MIL-STD-883, and with no intervening electrical measurements. If interim electrical tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of group A, subgroup 1, after burn-in, divided by the total number of devices submitted for burn-in in that lot shall be used to determined the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.
d. Constant acceleration test (method 2001 of MIL-STD-883) for case outline J shall be performed using, as a minimum, test condition D.

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4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections. (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 as follows:

a. Tests shall be as specified in table II herein.
b. Subgroups 4, 5, and 6 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883. Electrical tests shall be as specified in table II herein.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical tests shall be as specified in table II herein.
b. Subgroups 3 and 4 shall be added to the group C inspection requirements for class C devices and shall consist of the tests, conditions, and limits specified for subgroups 10 and 11 of group A.
c. Steady-state life test (method 1005 of MIL-STD-883) conditions:
   (1) Test condition D or E, using the circuit shown on figure 5, or equivalent.
   (2) T_A = 125°C minimum.
   (3) Test duration: 1,000 hours, except as permitted by appendix B of MIL-M-38510 and method 1005 of MIL-STD-883.
d. Constant acceleration test (method 2001 of MIL-STD-883) for case outline J shall be performed using, as a minimum, test condition D.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical tests shall be as specified in table II herein.

4.4.5 Inspection of packaging. Inspection of packaging shall be in accordance with MIL-M-38510.

4.5 Methods of inspection. Methods of inspection shall be as specified in appropriate tables and as specified herein.

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

5. PACKAGING

5.1 Packaging requirements. The requirements for the packaging of microcircuits shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Notes. The notes specified in MIL-M-38510 are applicable to this specification.

6.2 Intended use. Microcircuits conforming to this specification are intended for use for original equipment design applications and logistic support of existing equipment

6.3 Ordering data. The contract or order should specify the following:

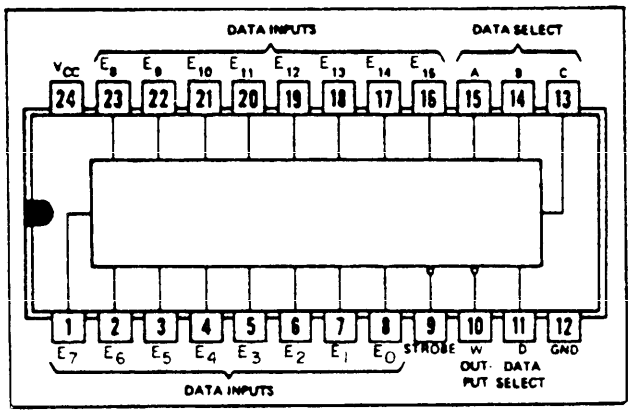
a. Complete part number (see 1.2).

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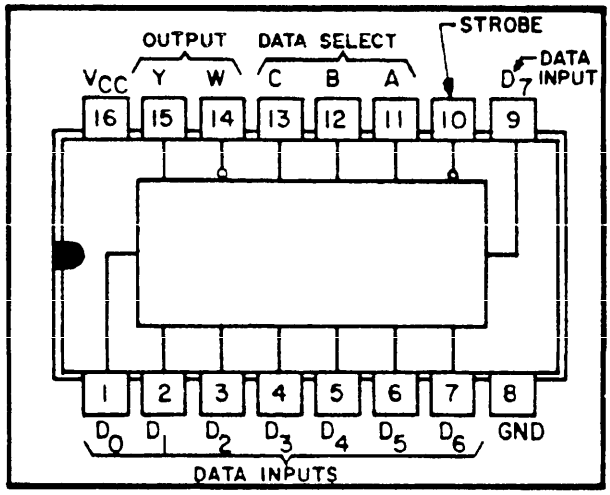
Device type 01

Cases J, K, and Z



Device type 02

Cases E and F



Device type 03

Cases E and F

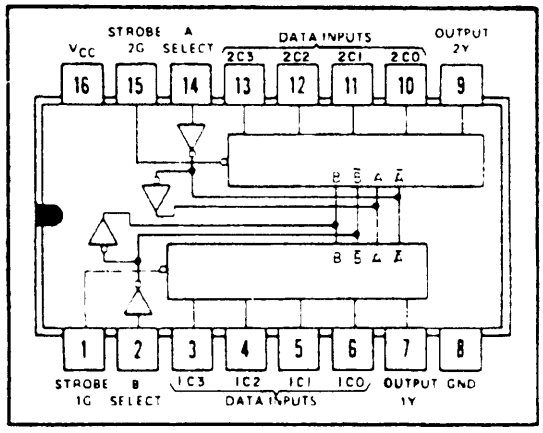
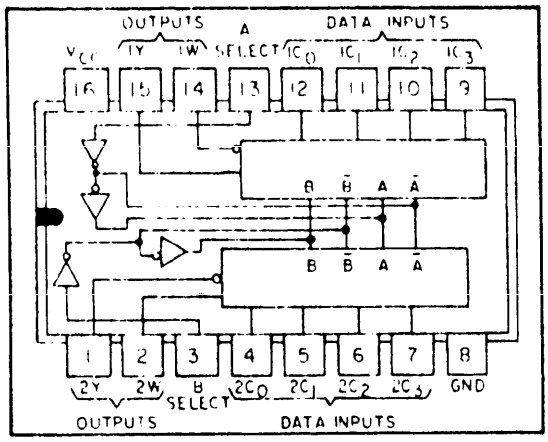


FIGURE 1. Terminal connections (top view).

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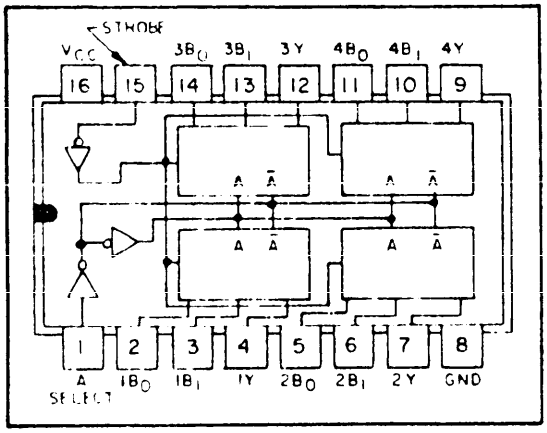
Device type 04

Cases E and F



Device type 05

Cases E and F



Device type 06

Cases E and F

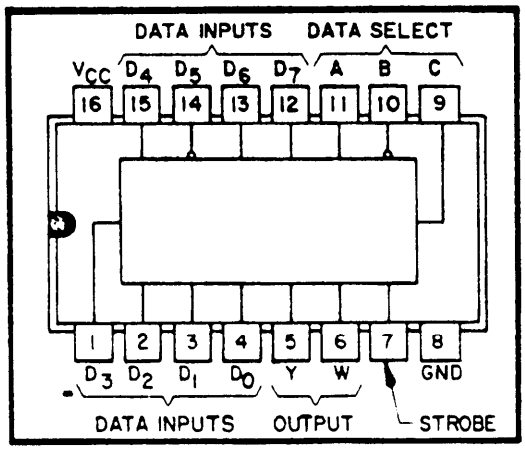


FIGURE 1. Terminal connections (top view) - Continued.

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Device type 01

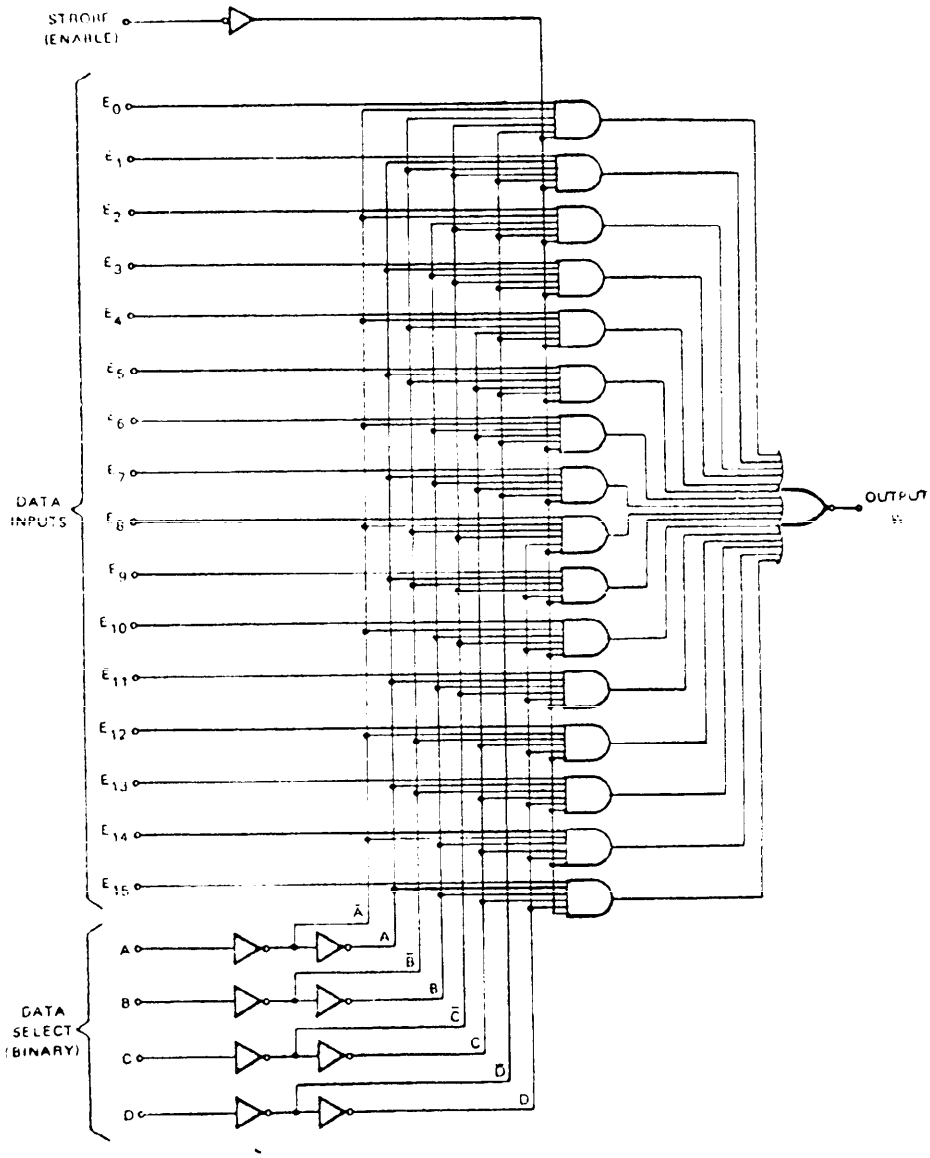


FIGURE 2. Logic diagrams.

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Device types 02 and 06

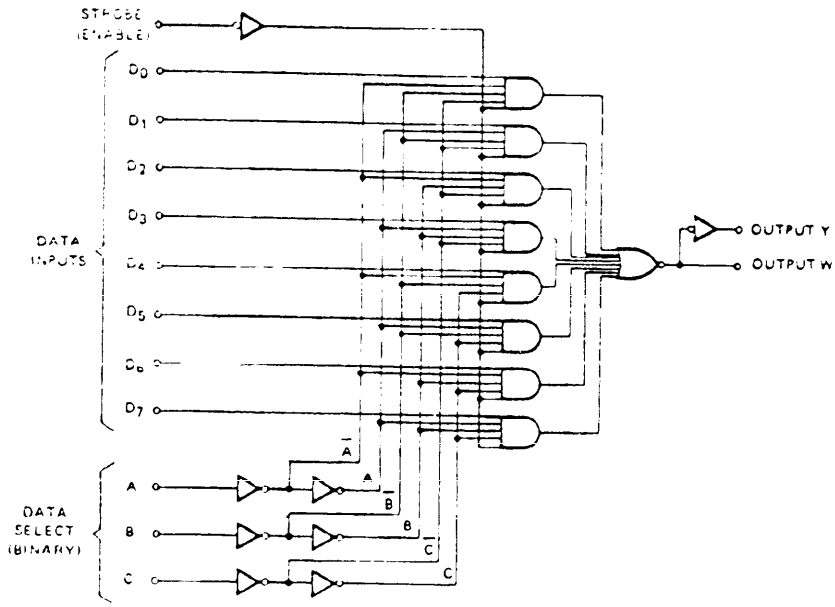


FIGURE 2. Logic diagrams - Continued.

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Device type 03

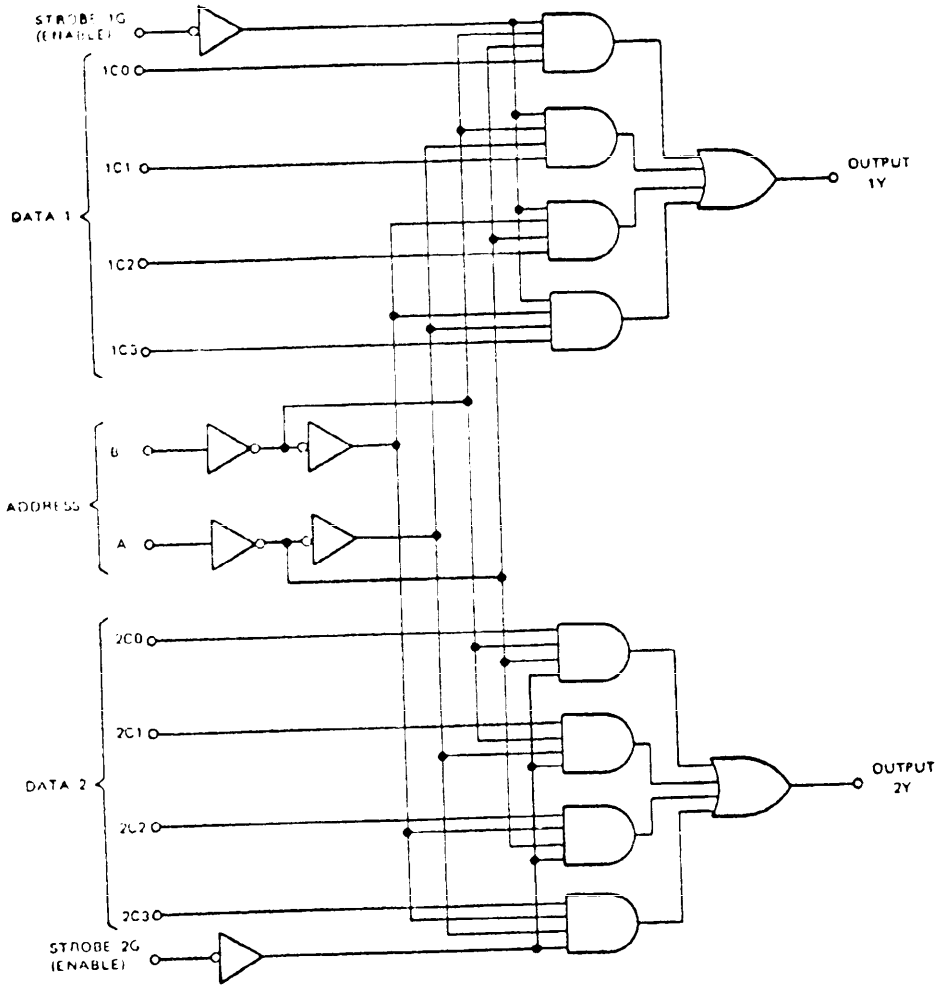
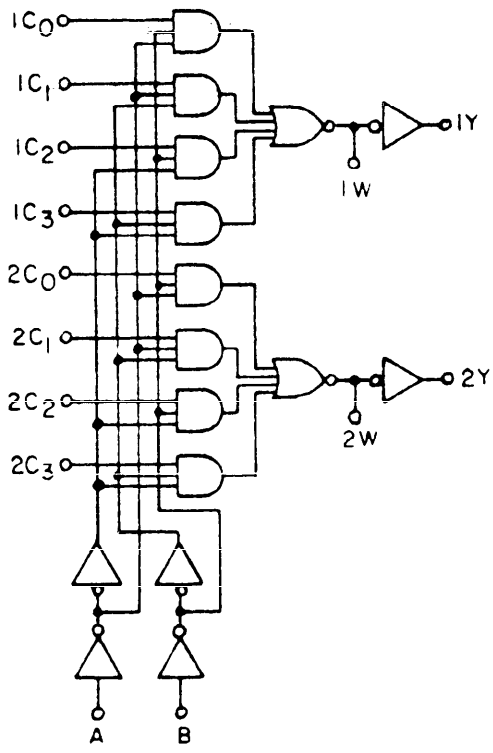


FIGURE 2. Logic diagrams - Continued.

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Device type 04



Device type 05

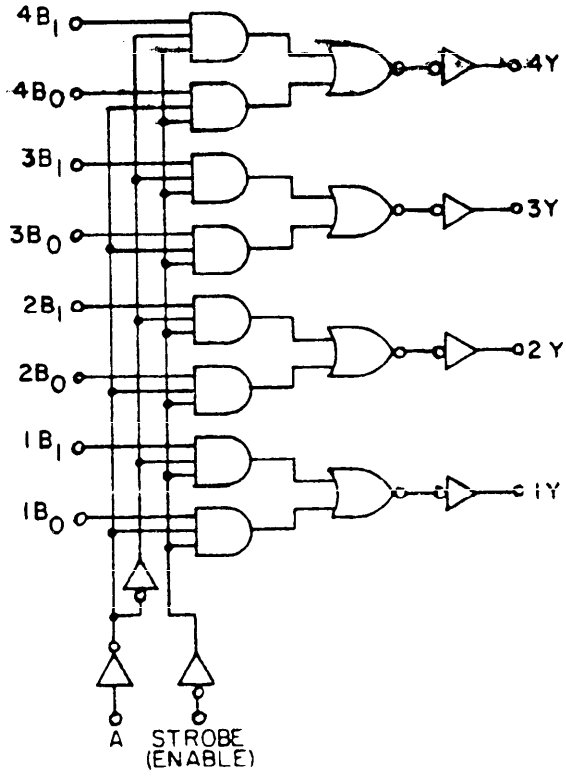


FIGURE 2. Logic diagrams - Continued.

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Device type 01

INPUTS																	OUTPUT
D C B A	STROBE	E_0	E_1	E_2	E_3	E_4	E_5	E_6	E_7	E_8	E_9	E_{10}	E_{11}	E_{12}	E_{13}	E_{14}	E_{15}	W
X X X X	H	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	H
L L L L	L	L	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	H
L L L L	L	H	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	L
L L L H	L	X	L	X	X	X	X	X	X	X	X	X	X	X	X	X	X	H
L L L H	L	X	H	X	X	X	X	X	X	X	X	X	X	X	X	X	X	L
L L H L	L	X	X	L	X	X	X	X	X	X	X	X	X	X	X	X	X	H
L L H L	L	X	X	H	X	X	X	X	X	X	X	X	X	X	X	X	X	L
L L H H	L	X	X	X	L	X	X	X	X	X	X	X	X	X	X	X	X	H
L L H H	L	X	X	X	H	X	X	X	X	X	X	X	X	X	X	X	X	L
L H L L	L	X	X	X	X	L	X	X	X	X	X	X	X	X	X	X	X	H
L H L L	L	X	X	X	X	H	X	X	X	X	X	X	X	X	X	X	X	L
L H L H	L	X	X	X	X	X	L	X	X	X	X	X	X	X	X	X	X	H
L H L H	L	X	X	X	X	X	H	X	X	X	X	X	X	X	X	X	X	L
L H H L	L	X	X	X	X	X	X	L	X	X	X	X	X	X	X	X	X	H
L H H L	L	X	X	X	X	X	X	H	X	X	X	X	X	X	X	X	X	L
L H H H	L	X	X	X	X	X	X	X	L	X	X	X	X	X	X	X	X	H
L H H H	L	X	X	X	X	X	X	X	H	X	X	X	X	X	X	X	X	L
H L L L	L	X	X	X	X	X	X	X	X	L	X	X	X	X	X	X	X	H
H L L L	L	X	X	X	X	X	X	X	X	H	X	X	X	X	X	X	X	L
H L L H	L	X	X	X	X	X	X	X	X	X	L	X	X	X	X	X	X	H
H L L H	L	X	X	X	X	X	X	X	X	X	H	X	X	X	X	X	X	L
H L H L	L	X	X	X	X	X	X	X	X	X	X	L	X	X	X	X	X	H
H L H L	L	X	X	X	X	X	X	X	X	X	X	H	X	X	X	X	X	L
H L H H	L	X	X	X	X	X	X	X	X	X	X	X	L	X	X	X	X	H
H L H H	L	X	X	X	X	X	X	X	X	X	X	X	H	X	X	X	X	L
H H L L	L	X	X	X	X	X	X	X	X	X	X	X	X	L	X	X	X	H
H H L L	L	X	X	X	X	X	X	X	X	X	X	X	X	H	X	X	X	L
H H L H	L	X	X	X	X	X	X	X	X	X	X	X	X	X	L	X	X	H
H H L H	L	X	X	X	X	X	X	X	X	X	X	X	X	X	H	X	X	L
H H H L	L	X	X	X	X	X	X	X	X	X	X	X	X	X	X	L	X	H
H H H L	L	X	X	X	X	X	X	X	X	X	X	X	X	X	X	H	X	L
H H H H	L	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	L	H
H H H H	L	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	H	L

When used to indicate an input condition, X = High logic level or low logic level.

FIGURE 3. Truth tables.

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Device types 02 and 06

INPUTS												OUTPUTS	
C	B	A	STROBE	D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7	Y	W
X	X	X	H	X	X	X	X	X	X	X	X	L	H
L	L	L	L	L	X	X	X	X	X	X	X	L	H
L	L	L	L	H	X	X	X	X	X	X	X	H	L
L	L	H	L	X	L	X	X	X	X	X	X	L	H
L	L	H	L	X	H	X	X	X	X	X	X	H	L
L	H	L	L	X	X	L	X	X	X	X	X	L	H
L	H	L	L	X	X	H	X	X	X	X	X	H	L
L	H	H	L	X	X	X	L	X	X	X	X	L	H
L	H	H	L	X	X	X	H	X	X	X	X	H	L
H	L	L	L	X	X	X	X	L	X	X	X	L	H
H	L	L	L	X	X	X	X	H	X	X	X	H	L
H	L	H	L	X	X	X	X	X	L	X	X	L	H
H	L	H	L	X	X	X	X	X	H	X	X	H	L
H	H	L	L	X	X	X	X	X	X	L	X	L	H
H	H	L	L	X	X	X	X	X	X	H	X	H	L
H	H	H	L	X	X	X	X	X	X	X	L	L	H
H	H	H	L	X	X	X	X	X	X	X	H	H	L

When used to indicate an input, X = Irrelevant.
H = High level, L = Low level.

ADDRESS INPUTS		DATA INPUTS				STROBE	OUTPUT
B	A	C0	C1	C2	C3	G	Y
X	X	X	X	X	X	H	L
L	L	L	X	X	X	L	L
L	L	H	X	X	X	L	H
L	H	X	L	X	X	L	L
L	H	X	H	X	X	L	H
H	L	X	X	L	X	L	L
H	L	X	X	H	X	L	H
H	H	X	X	X	L	L	L
H	H	X	X	X	H	L	H

Address inputs A and B are common to both sections.
H = high level, L = low level, X = irrelevant.

FIGURE 3. Truth tables - Continued.

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Device type 04

Address inputs		Data inputs				Outputs	
B	A	C_0	C_1	C_2	C_3	Y	W
L	L	L	X	X	X	L	H
L	L	H	X	X	X	H	L
L	H	X	L	X	X	L	H
L	H	X	H	X	X	H	L
H	L	X	X	L	X	L	H
H	L	X	X	H	X	H	L
H	H	X	X	X	L	L	H
H	H	X	X	X	H	H	L

Address inputs A and B are common to both sections.
H - High level, L = Low level, X = Irrelevant.

Device type 05

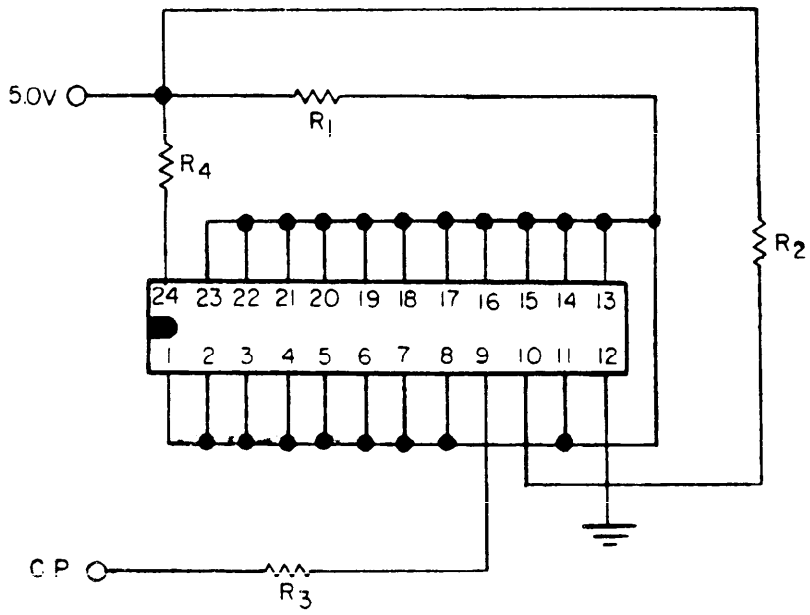
Strobe (enable)	Select input	Data inputs		Output
G	A	B_0	B_1	Y
H	X	X	X	L
L	H	X	L	L
L	H	X	H	H
L	L	L	X	L
L	L	H	X	H

Address A and strobe G are common to all sections.
H = High level, L = Low level, X = Irrelevant.

FIGURE 3. Truth tables - Continued.

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Device type 01

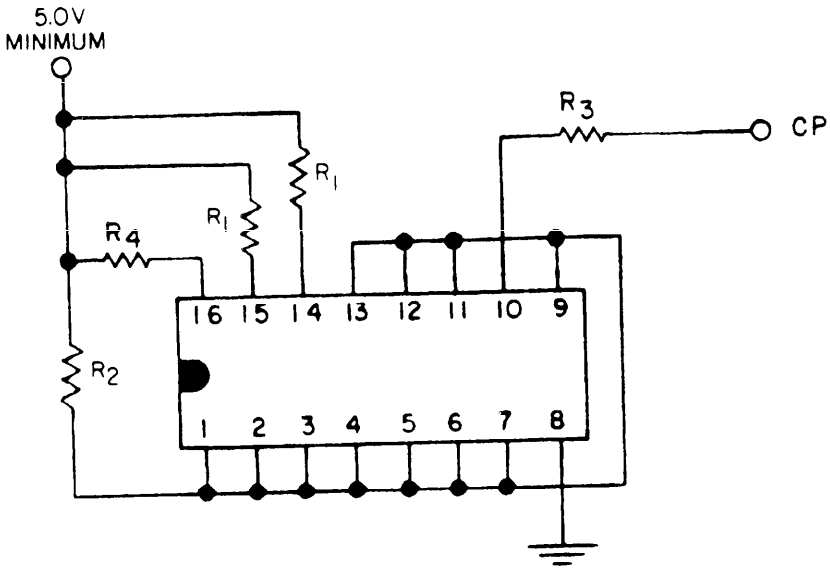


NOTES:
1. CP = 100 kHz ±5%; 50% duty cycle
   $-0.5\ V \leq V_{IL} \leq 0.8\ V$, $2.0\ V \leq V_{IH} \leq 5.5\ V$.
2. R1 = 300Ω ±5%.
3. R2 = 1 kΩ ±5%.
4. R3 = 27Ω maximum, R4 = 10Ω maximum.

FIGURE 4. Burn-in and life test circuits.

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Device type 02

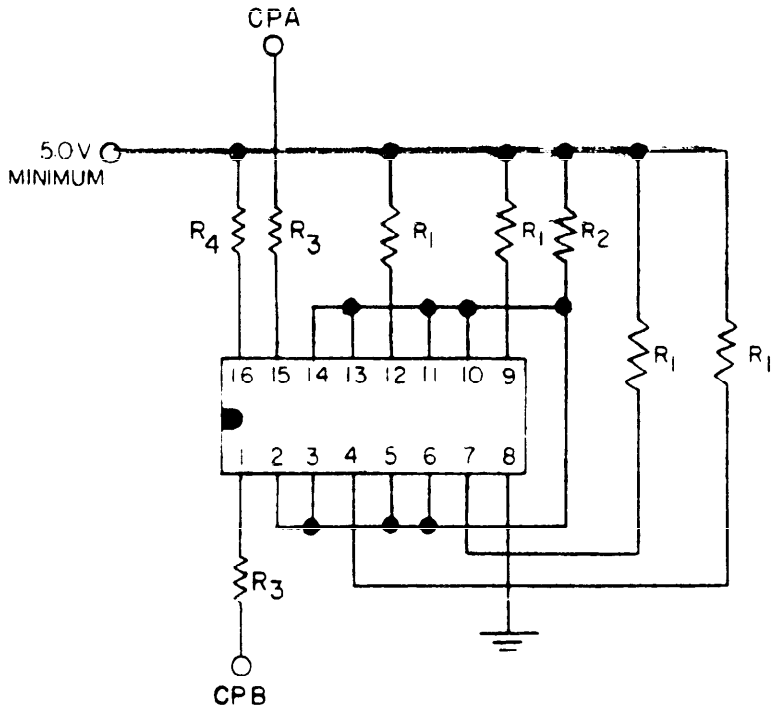


NOTES:
1. CP = 100 kHz ±5%; 50% duty cycle
   $-0.5\ V \leq V_{IL} \leq 0.8\ V$, $2.0\ V \leq V_{IH} \leq 5.5\ V$.
2. R1 = 300Ω ±5%.
3. R2 = 1 kΩ ±5%.
4. R3 = 27Ω maximum, R4 = 10Ω maximum.

FIGURE 4. Burn-in and life test circuits - Continued.

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Device type 03

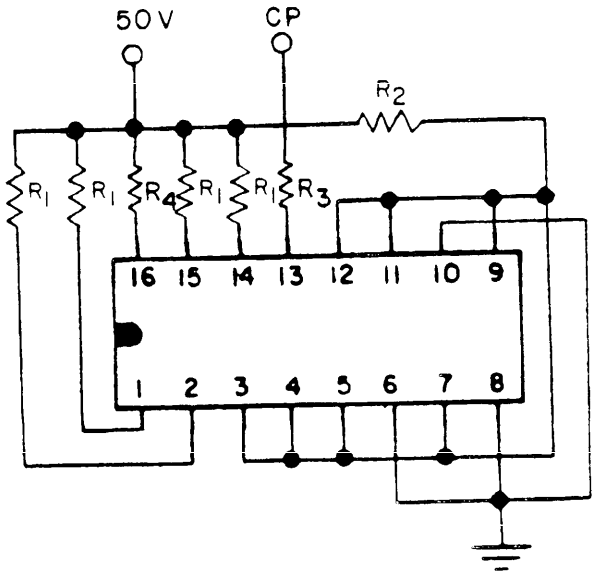


NOTES:
1. CPA = CPB = 100 kHz ±5%; 50% duty cycle
   $-0.5\ V \leq V_{IL} \leq 0.8\ V$, $2.0\ V \leq V_{IH} \leq 5.5\ V$.
2. R1 = 300Ω ±5%.
3. R2 = 1 kΩ ±5%.
4. R3 = 27Ω maximum, R4 = 10Ω maximum.

FIGURE 4. Burn-in and life test circuits - Continued.

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Device type 04

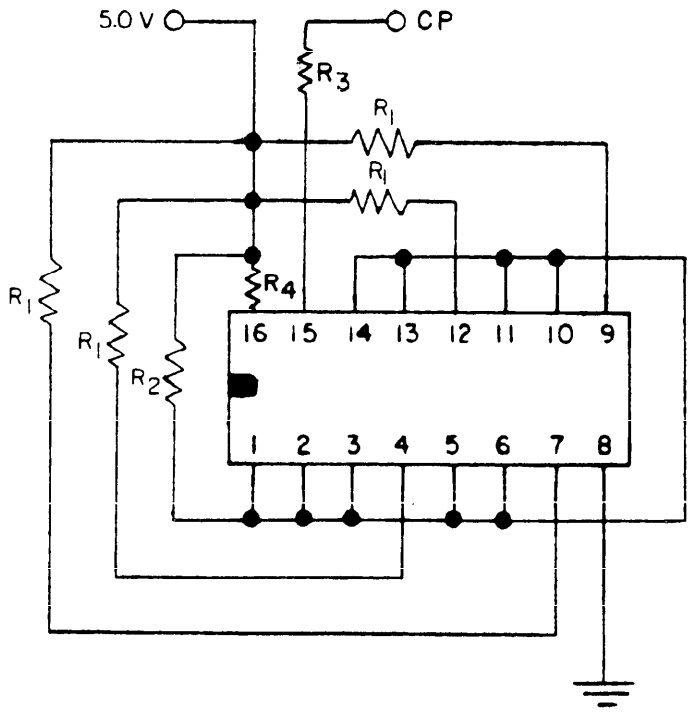


NOTES:
1. CP = 100 kHz ±5%; 50% duty cycle
   $-0.5\ V \leq V_{IL} \leq 0.8\ V$, $2.0\ V \leq V_{IH} \leq 5.5\ V$.
2. R1 = 300Ω ±5%.
3. R2 = 1 kΩ ±5%.
4. R3 = 27Ω maximum, R4 = 10Ω maximum.

FIGURE 4. Burn-in and life test circuits - Continued.

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Device type 05

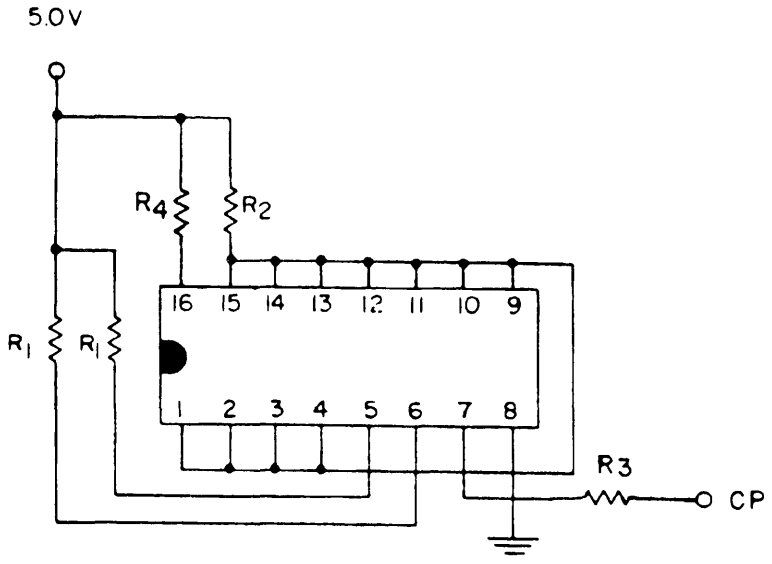


NOTES:
1. CP = 100 kHz ±5%; 50% duty cycle
   $-0.5\ V \leq V_{IL} \leq 0.8\ V$, $2.0\ V \leq V_{IH} \leq 5.5\ V$.
2. R1 = 300Ω ±5%.
3. R2 = 1 kΩ ±5%.
4. R3 = 27Ω maximum, R4 = 10Ω maximum.

FIGURE 4. Burn-in and life test circuits - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:51Z
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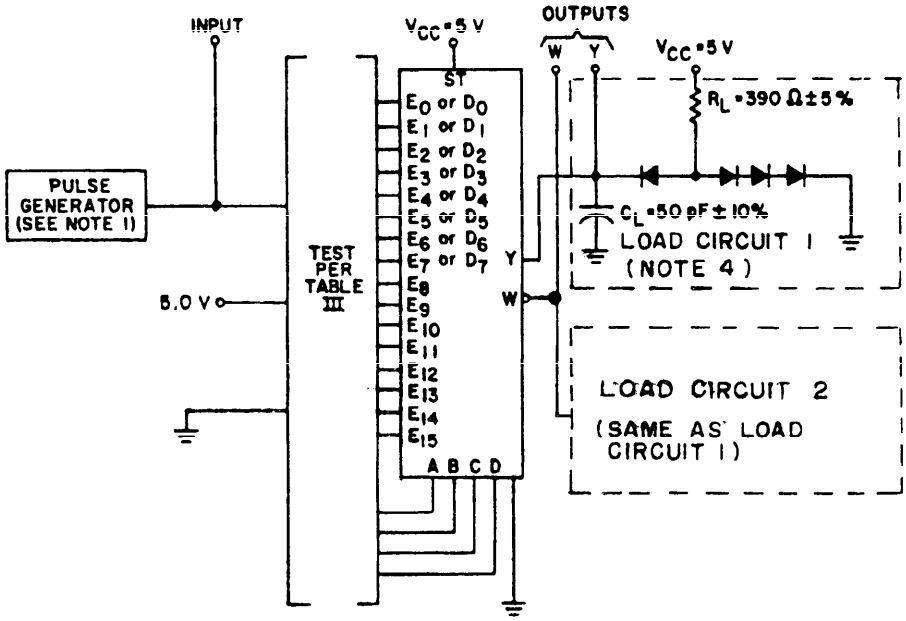
Device type 06

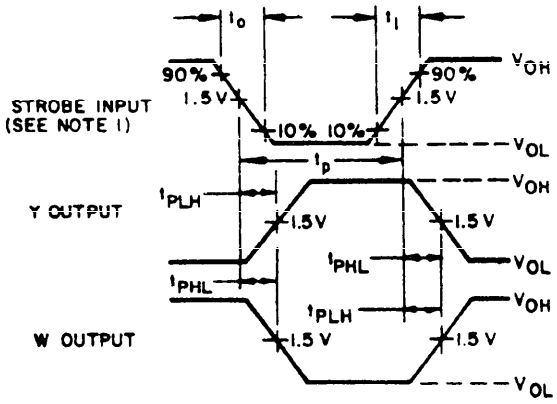


NOTES:
1. CP = 100 kHz ±5%; 50% duty cycle
   $-0.5\ V \leq V_{IL} \leq 0.8\ V$, $2.0\ V \leq V_{IH} \leq 5.5\ V$.
2. R1 = 300Ω ±5%.
3. R2 = 1 kΩ ±5%.
4. R3 = 27Ω maximum, R4 = 10Ω maximum.

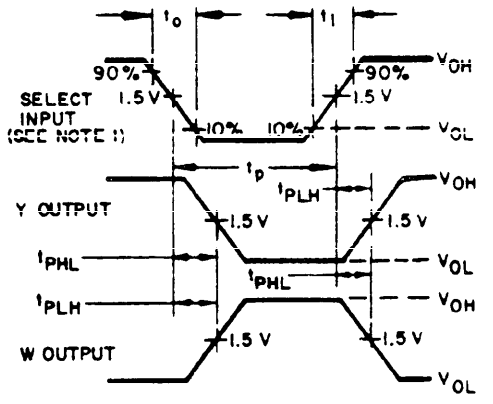
FIGURE 4. Burn-in and life test circuits - Continued.

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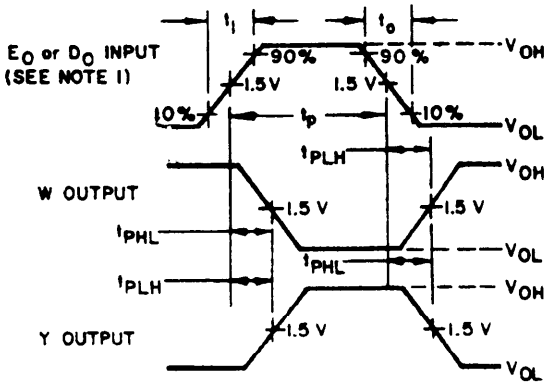




STROBE TO OUTPUT VOLTAGE WAVEFORMS



SELECT INPUT TO OUTPUT VOLTAGE WAVEFORMS

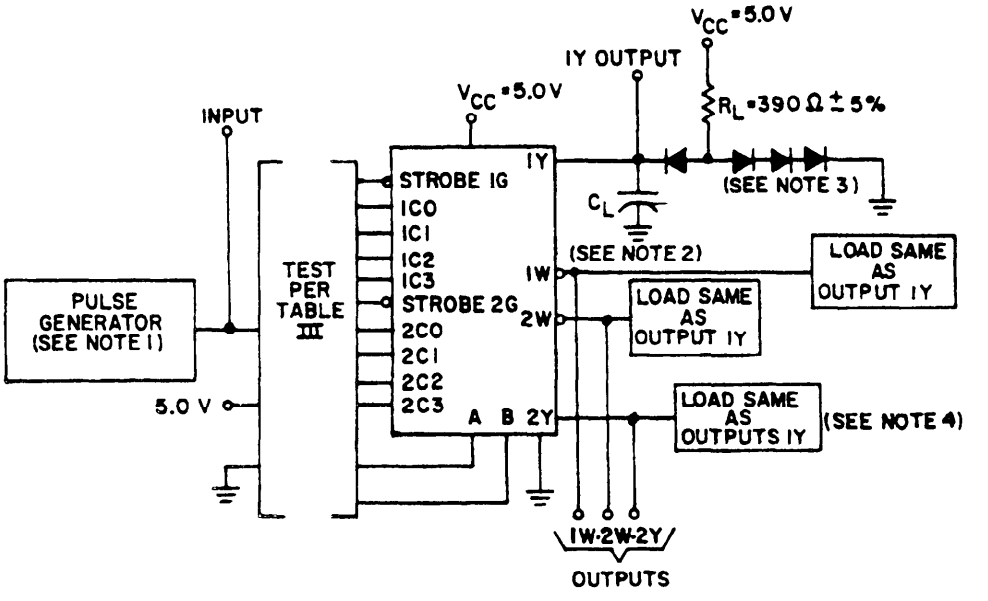


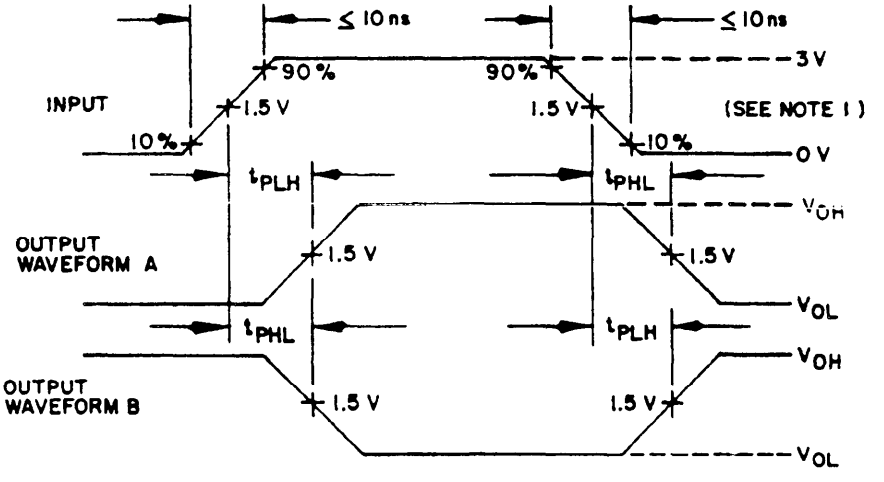
DATA INPUT TO OUTPUT VOLTAGE WAVEFORMS

NOTES:
1. The input pulse has the following characteristics: V_{OH} = 3 V, V_{OL} = 0 V, t_1 = t_0 = 10 ns, t_p = 500 ns, PRR = 1 MHz, duty cycle = 50%, and generator $Z_{out} \approx 50\,\Omega$.
2. C_L includes probe and jig capacitance.
3. All diodes are 1N3064 or equivalent.
4. Load circuits on a given output are only required where the specific test given in table III indicates "OUT" on that output. Load circuits may otherwise be omitted.

FIGURE 5. Switching test for device types 01, 02, and 06.

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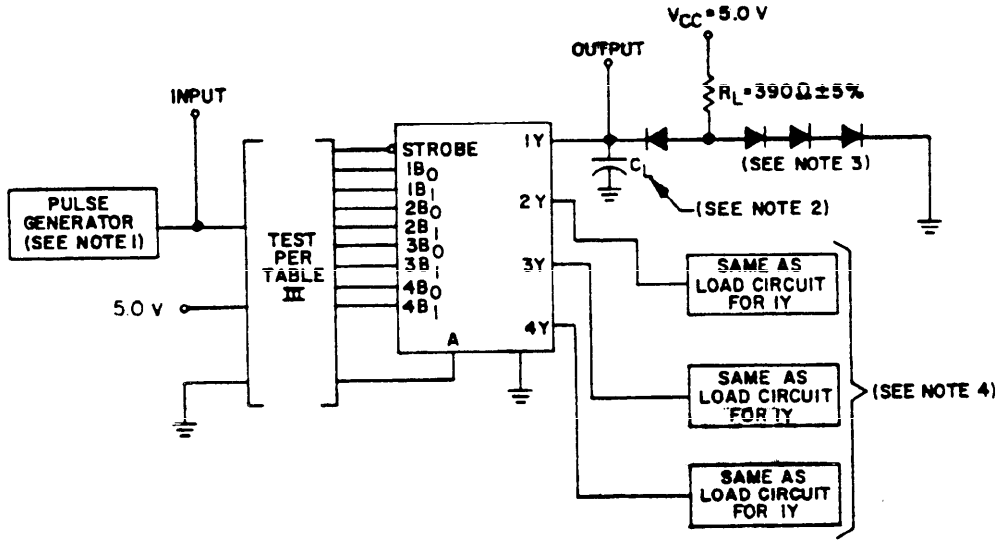
VOLTAGE WAVEFORMS

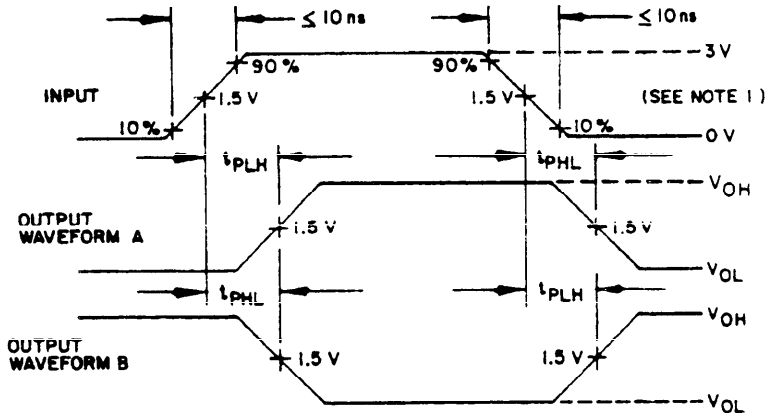
Switching time	Output waveform
CN to Y (types 03 and 04)	A
CN to W (type 04 only)	B
A or B to Y (types 03 and 04)	A
A or B to W (type 04 only)	B
G to Y (type 03 only)	B

NOTES:
1. The pulse generator has the following characteristics: PRR = 1 MHz, duty cycle = 50% and $Z_{out} \approx 50\Omega$.
2. C_L = 50 pF ±10% and includes probe and jig capacitance.
3. All diodes are 1N3064, or equivalent.
4. Load circuits on a given output are only required where the specific test given in table III indicates "OUT" on that output. Load circuits may otherwise be omitted.

FIGURE 6. Switching test for device types 03 and 04.

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VOLTAGE WAVEFORMS

Input	Output waveform
A to Y	A
B to Y	A
S to Y	B

NOTES:
1. The pulse generator has the following characteristics: PRR = 1 MHz, duty cycle = 50% and $Z_{out} \approx 50\Omega$.
2. C_L = 50 pF ±10% and includes probe and jig capacitance.
3. All diodes are 1N3064 or equivalent.
4. Load circuits on a given output are only required where the specific test given in table III indicates "OUT" on that output. Load circuits may otherwise be omitted.

FIGURE 7. Switching test for device type 05.

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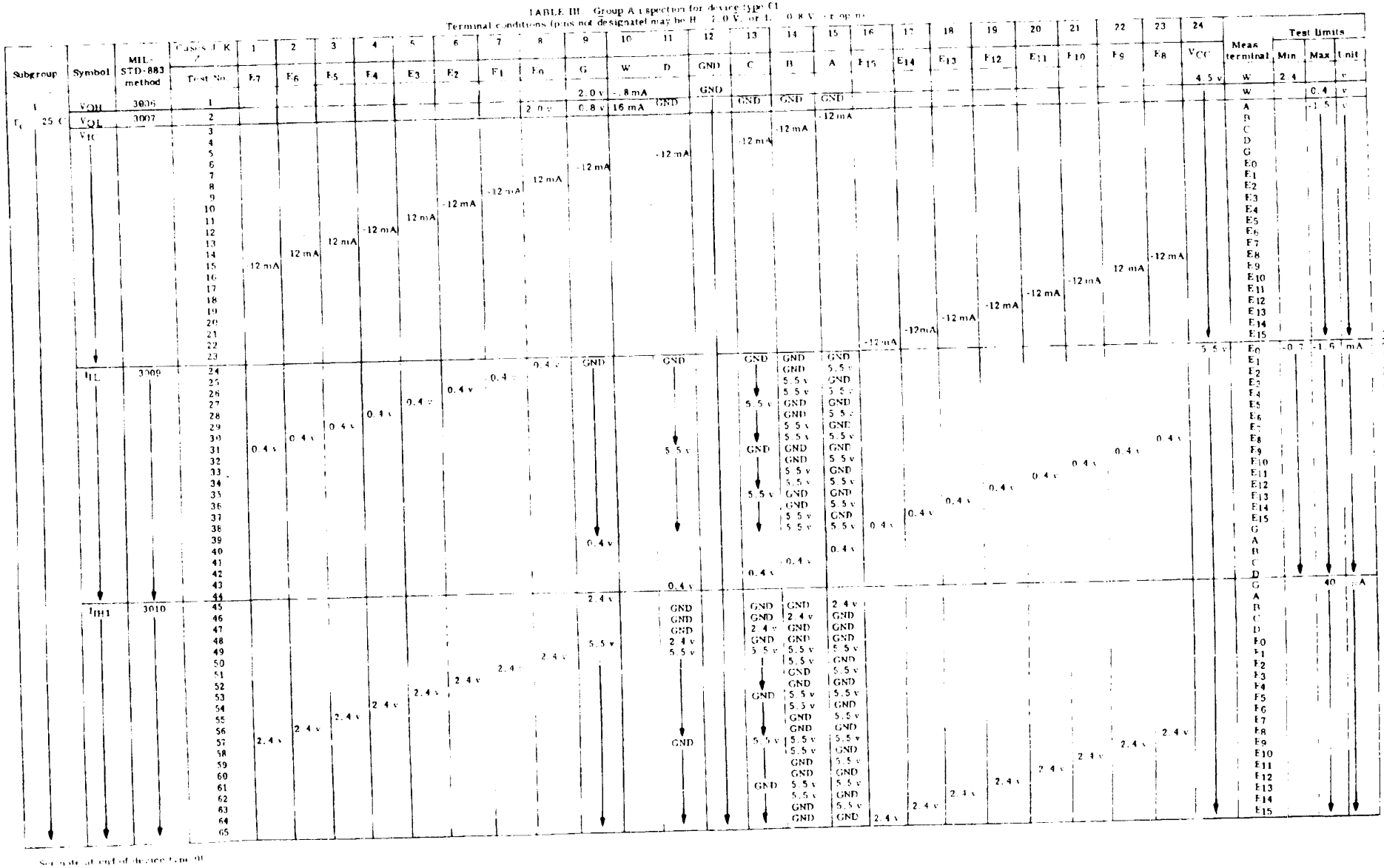


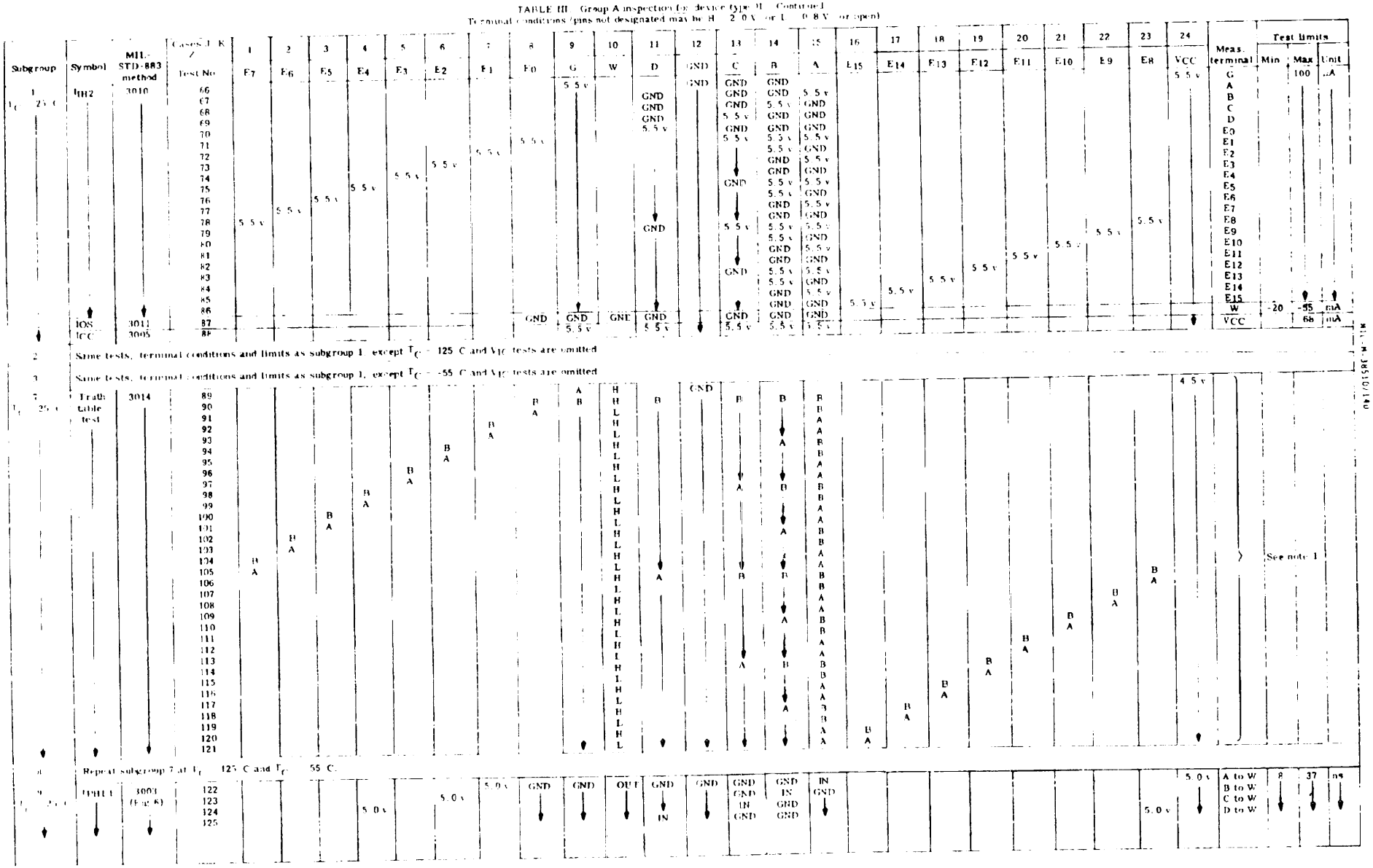
TABLE III. Group A inspection for device type 01

Terminal conditions (pins not designated may be H = 2.0 V, or L = 0.8 V, as specified)

Subgroup	Symbol	MIL-STD-883 method	Cases K	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Meas terminal	Test Limits Min	Test Limits Max	Test Limits Unit
			Test No.	E7	E6	E5	E4	E3	E2	E1	E0	G	W	D	GND	C	B	A	E15	E14	E13	E12	E11	E10	E9	E8	V_{CC}				
																											4.5 v	W	2.4		v
1	V_{OH}	3006	1								2.0 v	0.8 v	16 mA	GND	GND	GND	GND	-12 mA										W		0.4	v
T_A = 25°C	V_{OL}	3007	2																									A		-1.5	v
	V_{IC}		3													-12 mA	-12 mA											B			
			4																									C			
			5											-12 mA														D			
			6									-12 mA																G			
			7								12 mA																	E0			
			8							-12 mA																		E1			
			9						-12 mA																			E2			
			10					12 mA																				E3			
			11				-12 mA																					E4			
			12			12 mA																						E5			
			13		12 mA																							E6			
			14	-12 mA																								E7			
			15																									E8			
			16																						12 mA	-12 mA		E9			
			17																					-12 mA				E10			
			18																				-12 mA					E11			
			19																			-12 mA						E12			
			20																		-12 mA							E13			
			21																	-12 mA								E14			
			22																-12 mA									E15			
			23																									E0	-0.7	-1.6	mA
	I_{IL}	3009	24								0.4 v	GND	GND		GND	GND	GND	GND									5.5 v	E1			
			25													GND	5.5 v	5.5 v										E2			
			26							0.4 v						5.5 v	GND	GND										E3			
			27						0.4 v								GND	GND										E4			
			28					0.4 v									5.5 v	5.5 v										E5			
			29				0.4 v									GND	GND	GND										E6			
			30			0.4 v											5.5 v	5.5 v										E7			
			31		0.4 v											5.5 v	GND	GND										E8			
			32	0.4 v													GND	5.5 v								0.4 v		E9			
			33														5.5 v	GND							0.4 v			E10			
			34														GND	5.5 v						0.4 v				E11			
			35														5.5 v	GND					0.4 v					E12			
			36														GND	5.5 v				0.4 v						E13			
			37														5.5 v	GND			0.4 v							E14			
			38														5.5 v	5.5 v		0.4 v								E15			
			39															5.5 v										G			
			40									0.4 v						0.4 v										A			
			41														0.4 v											B			
			42													0.4 v												C			
			43											0.4 v														D			
			44																									G			
	I_{IH1}	3010	45									2.4 v			GND	GND	GND	GND										A		40	μA
			46												GND	GND	2.4 v	GND										B			
			47												GND	2.4 v	GND	GND										C			
			48												2.4 v	GND	GND	GND										D			
			49												GND	5.5 v	5.5 v	5.5 v										E0			
			50								2.4 v						5.5 v	GND										E1			
			51							2.4 v							GND	GND										E2			
			52						2.4 v								GND	GND										E3			
			53					2.4 v								GND	5.5 v	5.5 v										E4			
			54				2.4 v										5.5 v	GND										E5			
			55			2.4 v											GND	5.5 v										E6			
			56		2.4 v												GND	GND										E7			
			57	2.4 v												GND	5.5 v	5.5 v										E8			
			58														5.5 v	GND								2.4 v		E9			
			59														GND	GND							2.4 v			E10			
			60														GND	5.5 v						2.4 v				E11			
			61														5.5 v	GND					2.4 v					E12			
			62														GND	5.5 v				2.4 v						E13			
			63														5.5 v	GND			2.4 v							E14			
			64														GND	5.5 v		2.4 v								E15			
			65															GND										G			

See note at end of device type 01

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TABLE III. Group A inspection for device type 01 - Continued.

Terminal conditions (pins not designated may be H = 2.0 V or L = 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases J & 2 Test No.	1 E_7	2 E_6	3 E_5	4 E_4	5 E_3	6 E_2	7 E_1	8 E_0	9 G	0 W	11 D	12 GND	13 C	14 B	15 A	16 E_{15}	17 E_{14}	18 E_{13}	19 E_{12}	20 E_{11}	21 E_{10}	22 E_9	23 E_8	24 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
9 T_C = 25°C	$^tPLH1$	3003 (Fig. 6)	126						5.0 v	5.0 v	GND	GND	OUT	GND	GND	GND	GND	IN								5.0 v	A to W	8	39	ns
			127													GND	IN	GND									B to W			
			128				5.0 v									IN	GND											C to W			
			129											IN		GND	GND									5.0 v		D to W			
	$^tPHL2$		130								5.0 v	IN	OUT	GND		GND	GND	GND										G to W	6	34	ns
	$^tPLH2$		131								5.0 v	IN	OUT	GND		GND	GND	GND										G to W	6	28	ns
	$^tPHL3$		132								IN	GND					GND	GND										E_0 to W	3	18	ns
			133							IN							GND	5.0 v										E_1 to W			
			134						IN								5.0 v	GND										E_2 to W			
			135					IN									5.0 v	5.0 v										E_3 to W			
			136				IN									5.0 v	GND	GND										E_4 to W			
			137			IN											GND	5.0 v										E_5 to W			
			138		IN												5.0 v	GND										E_6 to W			
			139	IN													5.0 v	5.0 v										E_7 to W			
			140											5.0 v		5.0 v	GND	GND							IN			E_8 to W			
			141														GND	5.0 v						IN				E_9 to W			
			142														5.0 v	GND					IN					E_{10} to W			
			143														5.0 v	5.0 v				IN						E_{11} to W			
			144													5.0 v	GND	GND			IN							E_{12} to W			
			145														GND	5.0 v										E_{13} to W			
			146														5.0 v	GND		IN								E_{14} to W			
			147														5.0 v	5.0 v	IN									E_{15} to W			
	$^tPLH3$		148								IN					GND	GND	GND										E_0 to W	3	24	ns
			149							IN							GND	5.0 v										E_1 to W			
			150						IN								5.0 v	GND										E_2 to W			
			151					IN									5.0 v	5.0 v										E_3 to W			
			152				IN									5.0 v	GND	GND										E_4 to W			
			153			IN											GND	5.0 v										E_5 to W			
			154		IN												5.0 v	GND										E_6 to W			
			155	IN													5.0 v	5.0 v										E_7 to W			
			156											5.0 v		GND	GND	GND								IN		E_8 to W			
			157														GND	5.0 v						IN				E_9 to W			
			158														5.0 v	GND					IN					E_{10} to W			
			159														5.0 v	5.0 v				IN						E_{11} to W			
			160													5.0 v	GND	GND			IN							E_{12} to W			
			161														GND	5.0 v			IN							E_{13} to W			
			162														5.0 v	GND		IN								E_{14} to W			
			163														5.0 v	5.0 v	IN									E_{15} to W			
10 T_C = 125°C	$^tPHL1$		164							5.0 v	GND	GND	OUT	GND		GND	GND	IN										A to W	8	40	ns
			165						5.0 v							GND	IN	GND										B to W			
			166				5.0 v									IN	GND	GND										C to W			
			167											IN		GND	GND	GND								5.0 v		D to W			
	$^tPLH1$		168											GND		GND	GND	IN										A to W			
			169						5.0 v							GND	IN	GND										B to W		43	
			170				5.0 v									IN	GND	GND										C to W			
			171											IN		GND	GND	GND								5.0 v		D to W			
	$^tPHL2$		172								5.0 v	IN	OUT	GND		GND	GND	GND										G to W	6	37	ns
	$^tPLH2$		173								5.0 v	IN	OUT	GND		GND	GND	GND										G to W	6	32	ns
	$^tPHL3$		174								IN	GND	OUT	GND		GND	GND	GND										E_0 to W	3	23	ns
			175							IN							GND	5.0 v										E_1 to W			
			176						IN								5.0 v	GND										E_2 to W			
			177					IN									5.0 v	5.0 v										E_3 to W			
			178				IN									5.0 v	GND	GND										E_4 to W			
			179			IN											GND	5.0 v										E_5 to W			
			180		IN												5.0 v	GND										E_6 to W			
			181	IN													5.0 v	5.0 v										E_7 to W			
			182											5.0 v		GND	GND	GND								IN		E_8 to W			
			183														GND	5.0 v						IN				E_9 to W			
			184														5.0 v	GND					IN					E_{10} to W			
			185														5.0 v	5.0 v				IN						E_{11} to W			
			186													5.0 v	GND	GND			IN							E_{12} to W			
			187														GND	5.0 v			IN							E_{13} to W			
			188														5.0 v	GND		IN								E_{14} to W			
			189														5.0 v	5.0 v	IN									E_{15} to W			

See note at end of device type 01.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be H = 2.0 V or L = 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases J, K	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Meas. terminal	Test limits		
			Test No.	E_7	E_6	E_5	E_4	E_3	E_2	E_1	E_0	G	W	D	GND	C	B	A	E_{15}	E_{14}	E_{13}	E_{12}	E_{11}	E_{10}	E_9	E_8	V_{CC}		Min	Max	Unit
10 T_C = 125°C	t_{PLH3}	3003 (Fig. 6)	190								IN	GND	OUT	GND	GND	GND	GND	GND									5.0 v	E_0 to W	3	30	ns
			191							IN						GND	GND	5.0 v										E_1 to W			
			192						IN							5.0 v	5.0 v	GND										E_2 to W			
			193					IN								5.0 v	5.0 v	5.0 v										E_3 to W			
			194				IN									5.0 v	GND	GND										E_4 to W			
			195			IN											GND	5.0 v										E_5 to W			
			196		IN												5.0 v	GND										E_6 to W			
			197	IN													5.0 v	5.0 v								IN		E_7 to W			
			198											5.0 v		GND	GND	GND										E_8 to W			
			199														GND	5.0 v										E_9 to W			
			200														5.0 v	GND					IN					E_{10} to W			
			201														5.0 v	5.0 v						IN				E_{11} to W			
			202													5.0 v	GND	GND				IN						E_{12} to W			
			203														GND	5.0 v			IN							E_{13} to W			
			204														5.0 v	GND		IN								E_{14} to W			
			205														5.0 v	5.0 v	IN									E_{15} to W			
11	Same tests, terminal conditions and limits as subgroup 10 except T_C = -55°C																														

NOTE:

1. INPUTS: A = 2.0 V minimum, B = 0.8 V maximum.
   OUTPUTS: Output voltage shall be either
   (a) H = 2.4 volts minimum and L = 0.4 volt maximum when using a high speed checker double comparator, or
   (b) H ≥ 1.5 volts and L < 1.5 volts when using a high speed checker single comparator.
   (c) Only attributes data is required for subgroups 7 and 8.

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TABLE III. Group A inspection for device type 02.

Terminal conditions (pins not designated may be H = 2.0 V, or L = 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Test No.	1 D0	2 D1	3 D2	4 D3	5 D4	6 D5	7 D6	8 GND	9 D7	10 G	11 A	12 B	13 C	14 W	15 Y	16 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
1 T_C = 25°C	V_{OH}	3006	1	2.0 v							GND		0.8 v	0.8 v	0.8 v	0.8 v		-0.8mA	4.5 v	Y	2.4		V
	V_{OH}	3006	2										2.0 v	2.0 v	2.0 v	2.0 v	-0.8 mA			W	2.4		
	V_{OL}	3007	3										2.0 v	2.0 v	2.0 v	2.0 v		16 mA		Y		0.4	
	V_{OL}	3007	4	2.0 v									0.8 v	0.8 v	0.8 v	0.8 v	16 mA			W		0.4	
	V_{IC}		5	-12 mA																D0		-1.5	
			6		-12 mA															D1			
			7			-12 mA														D2			
			8				-12 mA													D3			
			9					-12 mA												D4			
			10						-12 mA											D5			
			11							-12 mA										D6			
			12									-12 mA								D7			
			13										-12 mA							G			
			14											-12 mA						A			
			15												-12 mA					B			
			16													-12 mA				C			
	I_{IL}	3009	17										0.4 v	5.5 v	5.5 v	5.5 v			5.5 v	G	-0.7	-1.6	mA
			18										GND	0.4 v	5.5 v					A			
			19											5.5 v	0.4 v					B			
			20											5.5 v	5.5 v	0.4 v				C			
			21	0.4 v										GND	GND	GND				D0			
			22		0.4 v									5.5 v	GND					D1			
			23			0.4 v								GND	5.5 v					D2			
			24				0.4 v							5.5 v	5.5 v					D3			
			25					0.4 v						GND	GND	5.5 v				D4			
			26						0.4 v					5.5 v	GND					D5			
			27							0.4 v				GND	5.5 v					D6			
			28									0.4 v		5.5 v	5.5 v					D7			
	I_{IH1}	3010	29										2.4 v	GND	GND	GND				G		40	µA
			30										5.5 v	2.4 v	GND	GND				A			
			31											GND	2.4 v	GND				B			
			32											GND	GND	2.4 v				C			
			33	2.4 v										5.5 v	5.5 v	5.5 v				D0			
			34		2.4 v									GND	5.5 v					D1			
			35			2.4 v								5.5 v	GND					D2			
			36				2.4 v							GND	GND					D3			
			37					2.4 v						5.5 v	5.5 v	GND				D4			
			38						2.4 v					GND	5.5 v					D5			
			39							2.4 v				5.5 v	GND					D6			
			40									2.4 v		GND	GND					D7			
	I_{IH2}	3010	41										5.5 v	GND	GND	GND				G		100	
			42										GND	5.5 v	GND					A			
			43											GND	5.5 v					B			
			44											GND	GND	5.5 v				C			
			45	5.5 v										5.5 v	5.5 v					D0			
			46		5.5 v									GND	5.5 v					D1			
			47			5.5 v								5.5 v	GND					D2			
			48				5.5 v							GND	GND					D3			
			49					5.5 v						5.5 v	5.5 v	GND				D4			
			50						5.5 v					GND	5.5 v					D5			
			51							5.5 v				5.5 v	GND					D6			
			52									5.5 v		GND	GND					D7			

See note at end of device type 02.

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Meas. terminal	Test limits		
			Test No.	D_0	D_1	D_2	D_3	D_4	D_5	D_6	GND	D_7	G	A	B	C	W	Y	V_{CC}		Min	Max	Unit
1 T_C = 25°C	I_{OS}	3011	53	GND	GND	GND	GND	GND	GND	GND	GND	GND	5.5 v	GND	GND	GND	GND		5.5 v	W	-20	-120	mA
	I_{OS}	3011	54	5.5 v	GND	GND	GND	GND	GND	GND	↓	GND	GND	GND	GND	GND		GND	↓	Y	-20	-120	mA
	I_{CC}	3005	55	5.5 v	GND	GND	GND	GND	GND	GND	↓	GND	GND	GND	GND	GND			↓	V_{CC}		48	mA
2	Same tests, terminal conditions and limits as subgroup 1, except T_C = 125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions and limits as subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						
7 T_C = 25°C	Truth table test		56	B	B	B	B	B	B	B	GND	B	A				H	L	4.5 v		See note 1		
			57	A	A	A	A	A	A	A	↓	A	B	B	B	B	H	L	↓				
			58										B	B	↓	↓	L	H					
			59										A	A			H	L					
			60										A	B	A		L	H					
			61										B	A	↓		H	L					
			62										B	B			L	H					
			63										A	A	B	B	H	L					
			64										A	B	↓	↓	L	H					
			65										B	B			H	L					
			66										B	A	A	A	L	H					
			67										A	B	↓	↓	H	L					
			68										A	A			L	H					
			69										B	A	B	B	H	L					
			70										B	B	↓	↓	L	H					
			71										A	A			H	L					
			72								↓		A	A	A	A	L	H	↓				
8	Repeat subgroup 7 at T_C = 125°C and T_C = -55°C.																						
9 T_C = 25°C	t_{PHL1}	3003 (Fig 6)	73	GND	5.0 v	5.0 v					GND		GND	IN	GND	GND	OUT		5.0 v	A to W	6	32	ns
	↓	↓	74	↓	↓	↓					↓		↓	GND	IN	GND	↓		↓	B to W	↓	↓	↓
			75					5.0 v						GND	GND	IN				C to W			
	t_{PLH1}		76		5.0 v									IN	GND	GND				A to W		29	
	↓		77			5.0 v								GND	IN	GND				B to W		↓	
			78					5.0 v						GND	GND	IN				C to W			
	t_{PHL2}		79		5.0 v									IN	GND	GND		OUT		A to Y		40	
	↓		80			5.0 v								GND	IN	GND				B to Y		↓	
			81					5.0 v						GND	GND	IN				C to Y			
	t_{PLH2}		82		5.0 v									IN	GND	GND				A to Y		39	
	↓		83	↓		5.0 v								GND	IN	GND		↓		B to Y		↓	
			84					5.0 v					↓	GND	GND	IN				C to Y	↓		
	t_{PHL3}		85	5.0 v									IN	GND	GND	GND	OUT			G to W	6	28	
	t_{PLH3}		86	↓											↓	↓	OUT			G to W	6	24	
	t_{PHL4}		87															OUT		G to Y	8	37	
	t_{PLH4}		88										↓	↓	↓	↓	↓	OUT		G to Y	8	35	
	t_{PHL5}		89	IN	IN	IN	IN	IN	IN	IN		IN	GND	GND	GND	GND	OUT			D_0 to W	3	20	
	↓	↓	90		↓	↓	↓	↓	↓	↓	↓	↓		5.0 v	GND	↓	↓		↓	D_1 to W	↓	↓	↓
			91											GND	5.0 v					D_2 to W			
			92											5.0 v	5.0 v					D_3 to W			
			93											GND	GND	5.0 v				D_4 to W			
			94											5.0 v	GND					D_5 to W			
			95											GND	5.0 v					D_6 to W			
			96								↓		↓	5.0 v	5.0 v	↓	↓		↓	D_7 to W	↓	↓	↓

See note at end of device type 02.

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TABLE III. Group A inspection for device type 02 - Continued.

Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Test No.	1 D_0	2 D_1	3 D_2	4 D_3	5 D_4	6 D_5	7 D_6	8 GND	9 D_7	10 G	11 A	12 B	13 C	14 W	15 Y	16 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
9 T_C = 25°C	t_{PLH5}	3003 (Fig 6)	97	IN							GND		GND	GND	GND	GND	OUT		5.0 v	D_0 to W	3	17	ns
			98		IN								5.0 v	GND						D_1 to W			
			99			IN							GND	5.0 v						D_2 to W			
			100				IN						5.0 v	5.0 v						D_3 to W			
			101					IN					GND	GND	5.0 v					D_4 to W			
			102						IN				5.0 v	GND						D_5 to W			
			103							IN			GND	5.0 v						D_6 to W			
			104									IN	5.0 v	5.0 v						D_7 to W			
	t_{PHL6}		105	IN									GND	GND	GND		OUT			D_0 to Y	6	29	
			106		IN								5.0 v	GND						D_1 to Y			
			107			IN							GND	5.0 v						D_2 to Y			
			108				IN						5.0 v	5.0 v						D_3 to Y			
			109					IN					GND	GND	5.0 v					D_4 to Y			
			110						IN				5.0 v	GND						D_5 to Y			
			111							IN			GND	5.0 v						D_6 to Y			
			112									IN	5.0 v	5.0 v						D_7 to Y			
	t_{PLH6}		113	IN									GND	GND	GND					D_0 to Y	6	29	
			114		IN								5.0 v	GND						D_1 to Y			
			115			IN							GND	5.0 v						D_2 to Y			
			116				IN						5.0 v	5.0 v						D_3 to Y			
			117					IN					GND	GND	5.0 v					D_4 to Y			
			118						IN				5.0 v	GND						D_5 to Y			
			119							IN			GND	5.0 v						D_6 to Y			
			120									IN	5.0 v	5.0 v						D_7 to Y			
10 T_C = 125°C	t_{PHL1}		121	GND	5.0 v									IN	GND	GND	OUT			A to W	6	40	
			122			5.0 v								GND	IN	GND				B to W			
			123					5.0 v						GND	GND	IN				C to W			
	t_{PLH1}		124		5.0 v									IN	GND	GND				A to W		38	
			125			5.0 v								GND	IN	GND				B to W			
			126					5.0 v						GND	GND	IN				C to W			
	t_{PHL2}		127		5.0 v									IN	GND	GND		OUT		A to Y	8	49	
			128			5.0 v								GND	IN	GND				B to Y			
			129					5.0 v						GND	GND	IN				C to Y			
	t_{PLH2}		130		5.0 v									IN	GND	GND				A to Y		45	
			131			5.0 v								GND	IN	GND				B to Y			
			132					5.0 v						GND	GND	IN				C to Y			
	t_{PHL3}		133	5.0 v									IN	GND	GND	GND	OUT			G to W	6	37	
	t_{PLH3}		134														OUT			G to W	6	35	
	t_{PHL4}		135															OUT		G to Y	8	46	
	t_{PLH4}		136															OUT		G to Y	8	42	
	t_{PHL5}		137	IN									GND	GND	GND	GND	OUT			D_0 to W	3	32	
			138		IN									5.0 v	GND					D_1 to W			
			139			IN								GND	5.0 v					D_2 to W			
			140				IN							5.0 v	5.0 v					D_3 to W			
			141					IN						GND	GND	5.0 v				D_4 to W			
			142						IN					5.0 v	GND					D_5 to W			
			143							IN				GND	5.0 v					D_6 to W			
			144									IN		5.0 v	5.0 v					D_7 to W			

See note at end of device type 02.

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be H = 2.0 V, or L = 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Meas. terminal	Test limits		
			Test No.	D_0	D_1	D_2	D_3	D_4	D_5	D_6	GND	D_7	G	A	B	C	W	Y	V_{CC}		Min	Max	Unit
10 T_C = 125°C	t_{PLH5}	3003 (Fig 6)	145	IN							GND		GND	GND	GND	GND	OUT		5.0 v	D_0 to W	3	26	ns
			146		IN									5.0 v	GND					D_1 to W			
			147			IN								GND	5.0 v					D_2 to W			
			148				IN							5.0 v	5.0 v					D_3 to W			
			149					IN						GND	GND	5.0 v				D_4 to W			
			150						IN					5.0 v	GND					D_5 to W			
			151							IN				GND	5.0 v					D_6 to W			
			152									IN		5.0 v	5.0 v					D_7 to W			
	t_{PHL6}		153	IN										GND	GND	GND		OUT		D_0 to Y	6	41	
			154		IN									5.0 v	GND					D_1 to Y			
			155			IN								GND	5.0 v					D_2 to Y			
			156				IN							5.0 v	5.0 v					D_3 to Y			
			157					IN						GND	GND	5.0 v				D_4 to Y			
			158						IN					5.0 v	GND					D_5 to Y			
			159							IN				GND	5.0 v					D_6 to Y			
			160									IN		5.0 v	5.0 v					D_7 to Y			
	t_{PLH6}		161	IN										GND	GND	GND				D_0 to Y		33	
			162		IN									5.0 v	GND					D_1 to Y			
			163			IN								GND	5.0 v					D_2 to Y			
			164				IN							5.0 v	5.0 v					D_3 to Y			
			165					IN						GND	GND	5.0 v				D_4 to Y			
			166						IN					5.0 v	GND					D_5 to Y			
			167							IN				GND	5.0 v					D_6 to Y			
			168									IN		5.0 v	5.0 v					D_7 to Y			
11	Same tests, terminal conditions and limits as subgroup 10, except T_A = -55° C.																						

NOTE:

1. INPUTS: A = 2.0 V, minimum, B = 0.8 V, maximum.

   OUTPUTS: Output voltages shall be either:

   (a) H = 2.4 volts minimum and L = 0.4 volt maximum when using a high speed checker double comparator, or
   (b) H $\geq$ 1.5 volts and L $<$ 1.5 volts when using a high speed checker single comparator.
   (c) Only attributes data is required for subgroups 7 and 8.

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TABLE III. Group A inspection for device type 03.

Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Test No.	1 1G	2 B	3 $1C_3$	4 $1C_2$	5 $1C_1$	6 $1C_0$	7 1Y	8 GND	9 2Y	10 $2C_0$	11 $2C_1$	12 $2C_2$	13 $2C_3$	14 A	15 2G	16 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
1 T_C = 25°C	V_{OH}	3006	1	0.8 v	0.8 v				2.0 v	-.8 mA	GND						0.8 v		4.5 v	1Y	2.4		V
	V_{OH}	3006	2		0.8 v							-.8 mA	2.0 v				0.8 v	0.8 v		2Y	2.4		
	V_{OL}	3007	3	2.0 v						16 mA										1Y		0.4	
	V_{OL}	3007	4									16 mA						2.0 v		2Y		0.4	
	V_{IC}		5		-12 mA				-12 mA								-12 mA			A		-1.5	
			6					-12 mA												B			
			7				-12 mA													$1C_0$			
			8			-12 mA														$1C_1$			
			9												-12 mA					$1C_2$			
			10	-12 mA									-12 mA			-12 mA				$1C_3$			
			11											-12 mA						1G			
			12																	$2C_0$			
			13																	$2C_1$			
			14																	$2C_2$			
			15															-12 mA		$2C_3$			
			16																5.5 v	2G			
	I_{IL}	3009	17														0.4 v			A	-0.7	-1.6	mA
			18		0.4 v															B			
			19	0.4 v														0.4 v		1G			
			20																	2G			
			21	GND	GND				0.4 v								GND			$1C_0$			
			22		GND												5.5 v			$1C_1$			
			23		5.5 v		0.4 v										GND			$1C_2$			
			24		5.5 v	0.4 v											5.5 v			$1C_3$			
			25		GND			0.4 v					0.4 v				GND	GND		$2C_0$			
			26		GND									0.4 v			5.5 v			$2C_1$			
			27		5.5 v										0.4 v		GND			$2C_2$			
			28		5.5 v											0.4 v	5.5 v			$2C_3$			
	I_{IH1}	3010	29		2.4 v												2.4 v			A		40	μA
			30																	B			
			31	2.4 v														2.4 v		1G			
			32														5.5 v			2G			
			33	5.5 v	5.5 v				2.4 v								GND			$1C_0$			
			34		5.5 v			2.4 v									5.5 v			$1C_1$			
			35		GND		2.4 v										GND			$1C_2$			
			36		GND	2.4 v											5.5 v	5.5 v		$1C_3$			
			37		5.5 v								2.4 v				GND			$2C_0$			
			38		5.5 v									2.4 v			5.5 v			$2C_1$			
			39		GND										2.4 v		GND			$2C_2$			
			40		GND											2.4 v				$2C_3$			

See note at end of device type 03.

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TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions (pins not designated may be H = 2.0 V, or L = 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Meas. terminal	Test limits		
			Test No.	1G	B	$1C_3$	$1C_2$	$1C_1$	$1C_0$	1Y	GND	2Y	$2C_0$	$2C_1$	$2C_2$	$2C_3$	A	2G	V_{CC}		Min	Max	Unit
1 T_C = 25°C	I_{IH2}	3010	41								GND						5.5 v		5.5 v	A		100	µA
			42		5.5 v															B			
			43	5.5 v																1G			
			44															5.5 v		2G			
			45	5.5 v	5.5 v				5.5 v								5.5 v			$1C_0$			
			46		5.5 v			5.5 v									GND			$1C_1$			
			47		GND		5.5 v										5.5 v			$1C_2$			
			48		GND	5.5 v											GND			$1C_3$			
			49		5.5 v								5.5 v				5.5 v	5.5 v		$2C_0$			
			50		5.5 v									5.5 v			GND			$2C_1$			
			51		GND										5.5 v		5.5 v			$2C_2$			
			52		GND											5.5 v	GND			$2C_3$			
	I_{OS}	3011	53	GND	GND	GND	GND	GND	5.5 v	GND							GND			1Y	-20	-55	mA
	I_{OS}	3011	54	GND	GND							GND	5.5 v	GND	GND	GND	GND	GND		2Y	-20	-55	mA
	I_{CC}	3005	55	GND	GND	GND	GND	GND	GND				GND	GND	GND	GND	GND	GND		V_{CC}		52	mA
2	Same tests, terminal conditions and limits as subgroup 1, except T_C = 125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions and limits as subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						
7 T_C = 25°C	Truth table test		56	A	B				B	L	GND	L	B				B	A	4.5 v		See note 1		
			57	B	B				A	L		L	A				B	B					
			58		B			B		H		H		B			A	A					
			59		B			A		L		L		A			A	B					
			60		A		B			H		H			B		B	B					
			61		A		A			L		L			A		B	A					
			62		A					H		H				B	A						
			63		A	B				L		L				A	A						
			64		A	A				H		H				A	A						
8	Repeat subgroup 7 at T_C = 125° C and T_C = -55° C.																						
9 T_C = 25°C	t_{PHL1}	3003 (Fig. 7)	65	GND	GND			IN	IN	OUT	GND						GND		5.0 v	$1C_0$ to 1Y	3	25	ns
			66		GND		IN										5.0 v			$1C_1$ to 1Y			
			67		5.0 v												GND			$1C_2$ to 1Y			
			68		5.0 v	IN											5.0 v			$1C_3$ to 1Y			
			69		GND							OUT	IN	IN			GND	GND		$2C_0$ to 2Y			
			70		GND										IN		5.0 v			$2C_1$ to 2Y			
			71		5.0 v											IN	GND			$2C_2$ to 2Y			
			72		5.0 v												5.0 v			$2C_3$ to 2Y			
	t_{PLH1}		73	GND	GND			IN	IN	OUT							GND			$1C_0$ to 1Y	3	24	
			74		GND		IN										5.0 v			$1C_1$ to 1Y			
			75		5.0 v												GND			$1C_2$ to 1Y			
			76		5.0 v	IN											5.0 v			$1C_3$ to 1Y			
			77		GND							OUT	IN	IN			GND	GND		$2C_0$ to 2Y			
			78		GND										IN		5.0 v			$2C_1$ to 2Y			
			79		5.0 v											IN	GND			$2C_2$ to 2Y			
			80		5.0 v												5.0 v			$2C_3$ to 2Y			
	t_{PHL2}		81	GND	GND			5.0 v	GND	OUT							IN			A to 1Y	6	36	
			82	GND	IN		5.0 v		GND	OUT							GND			B to 1Y			
			83		GND							OUT	GND	5.0 v			IN	GND		A to 2Y			
			84		IN							OUT	GND		5.0 v		GND	GND		B to 2Y			

See note at end of device type 03.

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TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions (pins not designated may be H = 2.0 V, or L = 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Meas. terminal	Test limits		
			Test No.	1G	B	$1C_3$	$1C_2$	$1C_1$	$1C_0$	1Y	GND	2Y	$2C_0$	$2C_1$	$2C_2$	$2C_3$	A	2G	V_{CC}		Min	Max	Unit
9	t_{PLH2}	3003	85	GND	GND			5.0 v	GND	OUT	GND						IN		5.0 v	A to 1Y	6	34	ns
T_C = 25°C		(Fig. 7)	86	GND	IN		5.0 v		GND	OUT							GND			B to 1Y			
			87		GND							OUT	GND	5.0 v			IN	GND		A to 2Y			
			88		IN							OUT	GND		5.0 v		GND	GND		B to 2Y			
	t_{PHL3}		89	IN	GND				5.0 v	OUT							GND			1G to 1Y		28	
	t_{PHL3}		90		GND							OUT	5.0 v				GND	IN		2G to 2Y		28	
	t_{PLH3}		91	IN	GND				5.0 v	OUT							GND			1G to 1Y		38	
	t_{PLH3}		92		GND							OUT	5.0 v				GND	IN		2G to 2Y		38	
10	t_{PHL1}		93	GND	GND				IN	OUT							GND			$1C_0$ to 1Y	3	29	
T_C = 125°C			94		GND			IN									5.0 v			$1C_1$ to 1Y			
			95		5.0 v		IN										GND			$1C_2$ to 1Y			
			96		5.0 v	IN											5.0 v			$1C_3$ to 1Y			
			97		GND							OUT	IN				GND	GND		$2C_0$ to 2Y			
			98		GND									IN			5.0 v			$2C_1$ to 2Y			
			99		5.0 v										IN		GND			$2C_2$ to 2Y			
			100		5.0 v											IN	5.0 v			$2C_3$ to 2Y			
	t_{PLH1}		101	GND	GND				IN	OUT							GND			$1C_0$ to 1Y		28	
			102		GND			IN									5.0 v			$1C_1$ to 1Y			
			103		5.0 v		IN										GND			$1C_2$ to 1Y			
			104		5.0 v	IN											5.0 v			$1C_3$ to 1Y			
			105		GND							OUT	IN				GND	GND		$2C_0$ to 2Y			
			106		GND									IN			5.0 v			$2C_1$ to 2Y			
			107		5.0 v										IN		GND			$2C_2$ to 2Y			
			108		5.0 v											IN	5.0 v			$2C_3$ to 2Y			
	t_{PHL2}		109	GND	GND			5.0 v	GND	OUT							IN			A to 1Y	6	44	
			110	GND	IN		5.0 v		GND	OUT							GND			B to 1Y			
			111		GND							OUT	GND	5.0 v			IN	GND		A to 2Y			
			112		IN							OUT	GND		5.0 v		GND	GND		B to 2Y			
	t_{PLH2}		113	GND	GND			5.0 v	GND	OUT							IN			A to 1Y		42	
			114	GND	IN		5.0 v		GND	OUT							GND			B to 1Y			
			115		GND							OUT	GND	5.0 v			IN	GND		A to 2Y			
			116		IN							OUT	GND		5.0 v		GND	GND		B to 2Y			
	t_{PHL3}		117	IN	GND				5.0 v	OUT							GND			1G to 1Y		32	
	t_{PHL3}		118		GND							OUT	5.0 v				GND	IN		2G to 2Y		32	
	t_{PLH3}		119	IN	GND				5.0 v	OUT							GND			1G to 1Y		42	
	t_{PLH3}		120		GND							OUT	5.0 v				GND	IN		2G to 2Y		42	
11	Same tests, terminal conditions and limits as subgroup 10, except T_C = -55°C.																						

NOTE:
1. INPUTS: A = 2.0 V, minimum, B = 0.8 V, maximum.
OUTPUTS: Output voltages shall be either:

(a) H = 2.4 volts minimum and L = 0.4 volt maximum when using a high speed checker double comparator, or
(b) H $\geq$ 1.5 volts and L $<$ 1.5 volts when using a high speed checker single comparator.
(c) Only attributes data is required for subgroups 7 and 8.

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TABLE III. Group A inspection for device type 04.

Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Meas. terminal	Test limits		
			Test No.	2Y	2W	B	$2C_0$	$2C_1$	$2C_2$	$2C_3$	GND	$1C_3$	$1C_2$	$1C_1$	$1C_0$	A	1W	1Y	V_{CC}		Min	Max	Unit
1 T_C = 25°C	V_{OH}	3006	1			0.8 v					GND				2.0 v	0.8 v		-0.8mA	4.5 v	1Y	2.4		V
			2												0.8 v		-0.8mA			1W			
			3	-0.8mA			2.0 v													2Y			
			4		-0.8mA		0.8 v													2W			
	V_{OL}	3007	5												2.0 v		16 mA			1W		0.4	
			6												0.8 v			16 mA		1Y			
			7		16 mA		2.0 v													2W			
			8	16 mA			0.8 v													2Y			
	V_{IC}		9			-12 mA									-12 mA	-12 mA				A		-1.5	
			10																	B			
			11																	$1C_0$			
			12																	$1C_1$			
			13									-12 mA	-12 mA	-12 mA						$1C_2$			
			14				-12 mA													$1C_3$			
			15					-12 mA												$2C_0$			
			16						-12 mA											$2C_1$			
			17							-12 mA										$2C_2$			
			18																	$2C_3$			
	I_{IL}	3009	19			0.4 v										0.4 v			5.5 v	A	-0.7	-1.6	mA
			20			GND									0.4 v	GND				B			
			21			GND										5.5 v				$1C_0$			
			22			5.5 v								0.4 v		GND				$1C_1$			
			23			5.5 v							0.4 v			5.5 v				$1C_2$			
			24			GND	0.4 v					0.4 v				GND				$1C_3$			
			25			GND		0.4 v								5.5 v				$2C_0$			
			26			5.5 v			0.4 v							GND				$2C_1$			
			27			5.5 v				0.4 v						5.5 v				$2C_2$			
			28																	$2C_3$			
	I_{IH1}	3010	29			2.4 v										2.4 v				A		40	μA
			30			5.5 v									2.4 v	5.5 v				B			
			31			5.5 v								2.4 v		GND				$1C_0$			
			32			GND							2.4 v			5.5 v				$1C_1$			
			33			GND						2.4 v				GND				$1C_2$			
			34			5.5 v	2.4 v									5.5 v				$1C_3$			
			35			5.5 v		2.4 v								GND				$2C_0$			
			36			GND			2.4 v							5.5 v				$2C_1$			
			37			GND				2.4 v						GND				$2C_2$			
			38																	$2C_3$			
	I_{IH2}		39			5.5 v										5.5 v				A		100	
			40												5.5 v	5.5 v				B			
			41											5.5 v		GND				$1C_0$			
			42										5.5 v			5.5 v				$1C_1$			
			43			GND						5.5 v				GND				$1C_2$			
			44			GND	5.5 v									5.5 v				$1C_3$			
			45			5.5 v		5.5 v								GND				$2C_0$			
			46			5.5 v			5.5 v							5.5 v				$2C_1$			
			47			GND				5.5 v						GND				$2C_2$			
			48																	$2C_3$			
	I_{OS}	3011	49									GND	GND	GND	GND		GND			1W	-20	-120	mA
			50									GND	GND	GND	5.5 v			GND		1Y			
			51		GND		GND	GND	GND	GND										2W			
			52	GND			5.5 v													2Y			
	I_{CC}	3005	53				5.5 v					GND	GND	GND	5.5 v					V_{CC}		45	
2	Same tests, terminal conditions and limits as subgroup 1, except T_C = 125° C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions and limits as subgroup 1, except T_C = -55° C and V_{IC} tests are omitted.																						

See note at end of device type 04.

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TABLE III. Group A inspection for device type 04 - Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Meas. terminal	Test limits		
			Test No.	2Y	2W	B	$2C_0$	$2C_1$	$2C_2$	$2C_3$	GND	$1C_3$	$1C_2$	$1C_1$	$1C_0$	A	1W	1Y	VCC		Min	Max	Unit
7 T_C = 25°C	Truth table test		54	L	H	B	B				GND				B	B	H	L	4.5 v				
			55	H	L		A								A	B	L	H					
			56	L	H			B						B		A	H	L					
			57	H	L			A						A		A	L	H					
			58	L	H	A			B				B			B	H	L			See note 1		
			59	H	L				A				A			B	I	H					
			60	L	H					B		B				A	H	I					
			61	H	L					A		A				A	L	H					
8	Repeat subgroup 7 at T_C = 125° C and T_C = -55° C.																						
9 T_C = 25°C	t_{PHL1}	3003 (Fig 7)	62			GND					GND				IN	GND		OUT	5.0 v	$1C_0$ to 1Y	3	29	ns
			63			GND								IN		5.0 v				$1C_1$ to 1Y			
			64			5.0 v							IN			GND				$1C_2$ to 1Y			
			65			5.0 v						IN				5.0 v				$1C_3$ to 1Y			
			66	OUT		GND	IN									GND				$2C_0$ to 2Y			
			67			GND		IN								5.0 v				$2C_1$ to 2Y			
			68			5.0 v			IN							GND				$2C_2$ to 2Y			
			69			5.0 v				IN						5.0 v				$2C_3$ to 2Y			
	t_{PLH1}		70			GND									IN	GND		OUT		$1C_0$ to 1Y		29	
			71			GND								IN		5.0 v				$1C_1$ to 1Y			
			72			5.0 v							IN			GND				$1C_2$ to 1Y			
			73			5.0 v						IN				5.0 v				$1C_3$ to 1Y			
			74	OUT		GND	IN									GND				$2C_0$ to 2Y			
			75			GND		IN								5.0 v				$2C_1$ to 2Y			
			76			5.0 v			IN							GND				$2C_2$ to 2Y			
			77			5.0 v				IN						5.0 v				$2C_3$ to 2Y			
	t_{PHL2}		78			GND									IN	GND	OUT			$1C_0$ to 1W		18	
			79			GND								IN		5.0 v				$1C_1$ to 1W			
			80			5.0 v							IN			GND				$1C_2$ to 1W			
			81			5.0 v						IN				5.0 v				$1C_3$ to 1W			
			82		OUT	GND	IN									GND				$2C_0$ to 2W			
			83			GND		IN								5.0 v				$2C_1$ to 2W			
			84			5.0 v			IN							GND				$2C_2$ to 2W			
			85			5.0 v				IN						5.0 v				$2C_3$ to 2W			
	t_{PLH2}		86			GND									IN	GND	OUT			$1C_0$ to 1W		17	
			87			GND								IN		5.0 v				$1C_1$ to 1W			
			88			5.0 v							IN			GND				$1C_2$ to 1W			
			89			5.0 v						IN				5.0 v				$1C_3$ to 1W			
			90		OUT	GND	IN									GND				$2C_0$ to 2W			
			91			GND		IN								5.0 v				$2C_1$ to 2W			
			92			5.0 v			IN							GND				$2C_2$ to 2W			
			93			5.0 v				IN						5.0 v				$2C_3$ to 2W			

See note at end of device type 04.

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TABLE III. Group A inspection for device type 04 - Continued.

Terminal conditions (pins not designated may be H ≥ 2.0V, or L ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Test No.	1 2Y	2 2W	3 B	4 $2C_0$	5 $2C_1$	6 $2C_2$	7 $2C_3$	8 GND	9 $1C_3$	10 $1C_2$	11 $1C_1$	12 $1C_0$	13 A	14 1W	15 1Y	16 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
9 T_C = 25°C	t_{PHL3}	3003 (Fig 7)	94	OUT		GND	GND	5.0V			GND			5.0V	GND	IN		OUT	5.0V	A to 1Y	6	37	ns
			95			GND										IN				A to 2Y			
			96			IN			5.0V				5.0V		GND	GND		OUT		B to 1Y			
			97	OUT		IN	GND								GND	GND				B to 2Y			
	t_{PLH3}		98			GND								5.0V	GND	IN		OUT		A to 1Y		37	
			99	OUT		GND	GND	5.0V								IN				A to 2Y			
			100			IN			5.0V				5.0V		GND	GND		OUT		B to 1Y			
			101	OUT		IN	GND								GND	GND				B to 2Y			
	t_{PHL4}		102			GND								5.0V	GND	IN	OUT			A to 1W		28	
			103		OUT	GND	GND	5.0V								IN				A to 2W			
			104			IN			5.0V				5.0V		GND	GND	OUT			B to 1W			
			105		OUT	IN	GND								GND	GND				B to 2W			
	t_{PLH4}		106			GND								5.0V	GND	IN	OUT			A to 1W		26	
			107		OUT	GND	GND	5.0V								IN				A to 2W			
			108			IN			5.0V				5.0V		GND	GND	OUT			B to 1W			
			109		OUT	IN	GND								GND	GND				B to 2W			
10 T_C = 125°C	t_{PHL1}		110			GND								IN	IN	GND		OUT		$1C_0$ to 1Y	3	41	
			111			GND										5.0V				$1C_1$ to 1Y			
			112			5.0V										GND				$1C_2$ to 1Y			
			113			5.0V						IN				5.0V				$1C_3$ to 1Y			
			114	OUT		GND	IN						IN			GND				$2C_0$ to 2Y			
			115			GND		IN								5.0V				$2C_1$ to 2Y			
			116			5.0V			IN							GND				$2C_2$ to 2Y			
			117			5.0V				IN						5.0V				$2C_3$ to 2Y			
	t_{PLH1}		118			GND								IN	IN	GND		OUT		$1C_0$ to 1Y		39	
			119			GND										5.0V				$1C_1$ to 1Y			
			120			5.0V										GND				$1C_2$ to 1Y			
			121			5.0V						IN				5.0V				$1C_3$ to 1Y			
			122	OUT		GND	IN						IN			GND				$2C_0$ to 2Y			
			123			GND		IN								5.0V				$2C_1$ to 2Y			
			124			5.0V			IN							GND				$2C_2$ to 2Y			
			125			5.0V				IN						5.0V				$2C_3$ to 2Y			
	t_{PHL2}		126			GND								IN	IN	GND	OUT			$1C_0$ to 1W		25	
			127			GND										5.0V				$1C_1$ to 1W			
			128			5.0V										GND				$1C_2$ to 1W			
			129			5.0V						IN				5.0V				$1C_3$ to 1W			
			130		OUT	GND	IN						IN			GND				$2C_0$ to 2W			
			131			GND		IN								5.0V				$2C_1$ to 2W			
			132			5.0V			IN							GND				$2C_2$ to 2W			
			133			5.0V				IN						5.0V				$2C_3$ to 2W			
	t_{PLH2}		134			GND								IN	IN	GND	OUT			$1C_0$ to 1W		24	
			135			GND										5.0V				$1C_1$ to 1W			
			136			5.0V										GND				$1C_2$ to 1W			
			137			5.0V						IN				5.0V				$1C_3$ to 1W			
			138		OUT	GND	IN						IN			GND				$2C_0$ to 2W			
			139			GND		IN								5.0V				$2C_1$ to 2W			
			140			5.0V			IN							GND				$2C_2$ to 2W			
			141			5.0V				IN						5.0V				$2C_3$ to 2W			

See note at end of device type 04.

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TABLE III. Group A inspection for device type 04 - Continued.

Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases F, S Test No.	1 2Y	2 2W	3 B	4 $2C_0$	5 $2C_1$	6 $2C_2$	7 $2C_3$	8 GND	9 $1C_3$	10 $1C_2$	11 $1C_1$	12 $1C_0$	13 A	14 1W	15 1Y	16 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
10 T_C = 25°C	t_{PHL3}	3003 (Fig 7)	142 143 144 145	OUT OUT		GND GND IN IN	GND GND	5.0V	5.0V		GND		5.0V	5.0V	GND GND	IN IN GND GND		OUT OUT	5.0V	A to 1Y A to 2Y B to 1Y B to 2Y	6	51	ns
	t_{PLH3}		146 147 148 149	OUT OUT		GND GND IN IN	GND GND	5.0V	5.0V				5.0V	5.0V	GND GND	IN IN GND GND		OUT OUT		A to 1Y A to 2Y B to 1Y B to 2Y			
	t_{PHL4}		150 151 152 153		OUT OUT	GND GND IN IN	GND GND	5.0V	5.0V				5.0V	5.0V	GND GND	IN IN GND GND	OUT OUT			A to 1W A to 2W B to 1W B to 2W		39	
	t_{PLH4}		154 155 156 157		OUT OUT	GND GND IN IN	GND GND	5.0V	5.0V				5.0V	5.0V	GND GND	IN IN GND GND	OUT OUT			A to 1W A to 2W B to 1W B to 2W		34	
11	Same tests, terminal conditions and limits as subgroup 10, except T_C = -55°C.																						

NOTE:

1. INPUTS: A = 2.0 V, minimum; B = 0.8 V, maximum.
   OUTPUTS: Output voltages shall be either:

   (a) H = 2.4 volts minimum and L = 0.4 volt maximum when using a high speed checker double comparator, or
   (b) H = 1.5 volts and L = 1.5 volts when using a high speed checker single comparator.
   (c) Only attributes data is required for subgroups 7 and 8.

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TABLE III. Group A inspection for device type 05.

Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Meas. terminal	Test limits		
			Test No.	A	$1B_0$	$1B_1$	1Y	$2B_0$	$2B_1$	2Y	GND	4Y	$4B_1$	$4B_0$	3Y	$3B_1$	$3B_0$	G	V_{CC}		Min	Max	Unit
1 T_C = 25°C	V_{OH}	3006	1 2 3 4	2.0 v		2.0 v	-.8 mA		2.0 v	-.8 mA	GND	-.8 mA	2.0 v		-.8 mA	2.0 v		0.8 v	4.5 v	1Y 2Y 3Y 4Y	2.4		v
	V_{OL}	3007	5 6 7 8				16 mA			16 mA		16 mA			16 mA			2.0 v		1Y 2Y 3Y 4Y		0.4	
	V_{IC}		9 10 11 12 13 14 15 16 17 18	-12mA	-12mA	-12mA		-12mA	-12mA				-12mA	-12mA		-12mA	-12mA	-12mA		A $1B_0$ $1B_1$ $2B_0$ $2B_1$ $4B_1$ $4B_0$ $3B_1$ $3B_0$ G		-1.5	
	I_{IL}	3009	19 20 21 22 23 24 25 26 27 28	0.4 v GND 5.5 v GND 5.5 v 5.5 v GND 5.5 v GND	0.4 v	0.4 v		0.4 v	0.4 v				0.4 v	0.4 v		0.4 v	0.4 v	0.4 v	0.4 v GND	5.5 v	G A $1B_0$ $1B_1$ $2B_0$ $2B_1$ $4B_1$ $4B_0$ $3B_1$ $3B_0$	-0.7	-1.6	mA
	I_{IH1}	3010	29 30 31 32 33 34 35 36 37 38	2.4 v 5.5 v GND 5.5 v GND GND 5.5 v GND 5.5 v	2.4 v	2.4 v		2.4 v	2.4 v				2.4 v	2.4 v		2.4 v	2.4 v	2.4 v	2.4 v 5.5 v		G A $1B_0$ $1B_1$ $2B_0$ $2B_1$ $4B_1$ $4B_0$ $3B_1$ $3B_0$		40	µA

See note at end of device type 05.

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TABLE III. Group A inspection for device type 05 - Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Meas. terminal	Test limits		
			Test No.	A	$1B_0$	$1B_1$	1Y	$2B_0$	$2B_1$	2Y	GND	4Y	$4B_1$	$4B_0$	3Y	$3B_1$	$3B_0$	G	V_{CC}		Min	Max	Unit
1 T_C = 25°C	I_{IH2}	3010	39 40 41 42 43 44 45 46 47 48	5.5 v 5.5 v GND 5.5 v GND GND 5.5 v GND 5.5 v	5.5 v	5.5 v		5.5 v	5.5 v		GND		5.5 v	5.5 v		5.5 v	5.5 v	5.5 v	5.5 v	G A $1B_0$ $1B_1$ $2B_0$ $2B_1$ $4B_1$ $4B_0$ $3B_1$ $3B_0$		100	µA
	I_{OS}	3011	49 50 51 52	5.5 v	5.5 v	5.5 v	GND	5.5 v	5.5 v	GND		GND	5.5 v	5.5 v	GND	5.5 v	5.5 v	GND		1Y 2Y 4Y 3Y	-20	-120	mA
	I_{CC}	3005	53	GND	GND	GND		GND	GND				GND	GND		GND	GND	GND		V_{CC}		50	
2	Same tests, terminal conditions and limits as subgroup 1, except T_C = 125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions and limits as subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						
7 T_C = 25°C	Truth table test		54 55 56 57 58	A A B B	B A	B A	L L H L H	B A	B A	L L H L H	GND	L L H L H	B A	B A	L L H L H	B A	B A	A B	4.5 v	See note 1			
8	Repeat subgroup 7 at T_C = 125° C and T_C = -55° C.																						
9 T_C = 25°C	t_{PHL1}	3003 (Fig 8)	59 60 61 62	IN	GND	5.0 v	OUT	GND	5.0 v	OUT	GND	OUT	5.0 v	GND	OUT	5.0 v	GND	GND	5.0 v	A to 1Y A to 2Y A to 3Y A to 4Y	6	30	ns
	t_{PLH1}		63 64 65 66		GND	5.0 v	OUT	GND	5.0 v	OUT		OUT	5.0 v	GND	OUT	5.0 v	GND			A to 1Y A to 2Y A to 3Y A to 4Y		27	

See note at end of device type 05.

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TABLE III. Group A inspection for device type 05 - Continued.

Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Meas. terminal	Test limits		
			Test No.	A	$1B_0$	$1B_1$	1Y	$2B_0$	$2B_1$	2Y	GND	4Y	$4B_1$	$4B_0$	3Y	$3B_1$	$3B_0$	G	V_{CC}		Min	Max	Unit
9	t_{PHL2}	3003	67	5.0 v		5.0 v	OUT		5.0 v	OUT	GND							IN	5.0 v	G to 1Y	3	28	ns
T_C = 25°C		(Fig. 8)	68												OUT	5.0 v				G to 2Y			
			69									OUT	5.0 v							G to 3Y			
			70																	G to 4Y			
	t_{PLH2}		71			5.0 v	OUT		5.0 v	OUT										G to 1Y		23	
			72												OUT	5.0 v				G to 2Y			
			73									OUT	5.0 v							G to 3Y			
			74																	G to 4Y			
	t_{PHL3}		75	GND	IN		OUT											GND		$1B_0$ to 1Y		20	
			76	5.0 v		IN	OUT	IN	IN	OUT					OUT	IN	IN			$1B_1$ to 1Y			
			77	GND						OUT					OUT					$2B_0$ to 2Y			
			78	5.0 v																$2B_1$ to 2Y			
			79	GND																$3B_0$ to 3Y			
			80	5.0 v																$3B_1$ to 3Y			
			81	GND																$4B_0$ to 4Y			
			82	5.0 v																$4B_1$ to 4Y			
	t_{PLH3}		83	GND	IN		OUT													$1B_0$ to 1Y		20	
			84	5.0 v		IN	OUT	IN	IN	OUT					OUT	IN	IN			$1B_1$ to 1Y			
			85	GND						OUT					OUT					$2B_0$ to 2Y			
			86	5.0 v																$2B_1$ to 2Y			
			87	GND																$3B_0$ to 3Y			
			88	5.0 v																$3B_1$ to 3Y			
			89	GND																$4B_0$ to 4Y			
			90	5.0 v																$4B_1$ to 4Y			
10	t_{PHL1}	3003	91	IN	GND	5.0 v	OUT	GND	5.0 v	OUT										A to 1Y	6	49	
T_C = 125°C		(Fig. 8)	92												OUT	5.0 v	GND			A to 2Y			
			93									OUT	5.0 v	GND						A to 3Y			
			94																	A to 4Y			
	t_{PLH1}		95		GND	5.0 v	OUT	GND	5.0 v	OUT										A to 1Y		41	
			96												OUT	5.0 v	GND			A to 2Y			
			97									OUT	5.0 v	GND						A to 3Y			
			98																	A to 4Y			
	t_{PHL2}		99	5.0 v		5.0 v	OUT		5.0 v	OUT								IN		G to 1Y	3	39	
			100												OUT	5.0 v				G to 2Y			
			101									OUT	5.0 v							G to 3Y			
			102																	G to 4Y			
	t_{PLH2}		103			5.0 v	OUT		5.0 v	OUT										G to 1Y		33	
			104												OUT	5.0 v				G to 2Y			
			105									OUT	5.0 v							G to 3Y			
			106																	G to 4Y			
	t_{PHL3}		107	GND	IN		OUT											GND		$1B_0$ to 1Y		25	
			108	5.0 v		IN	OUT	IN	IN	OUT					OUT	IN	IN			$1B_1$ to 1Y			
			109	GND						OUT					OUT					$2B_0$ to 2Y			
			110	5.0 v																$2B_1$ to 2Y			
			111	GND																$3B_0$ to 3Y			
			112	5.0 v																$3B_1$ to 3Y			
			113	GND																$4B_0$ to 4Y			
			114	5.0 v																$4B_1$ to 4Y			

See note at end of device type 05.

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TABLE III. Group A inspection for device type 05 - Continued.

Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Meas. terminal	Test limits		
			Test No.	A	$1B_0$	$1B_1$	1Y	$2B_0$	$2B_1$	2Y	GND	4Y	$4B_1$	$4B_0$	3Y	$3B_1$	$3B_0$	G	V_{CC}		Min	Max	Unit
10 T_C 125°C	t_{PLH3}	3003 (Fig 8)	115	GND	IN		OUT				GND							GND	5.0 v	$1B_0$ to 1Y	3	35	ns
			116	5.0 v		IN	OUT													$1B_1$ to 1Y			
			117	GND				IN		OUT										$2B_0$ to 2Y			
			118	5.0 v					IN	OUT										$2B_1$ to 2Y			
			119	GND											OUT					$3B_0$ to 3Y			
			120	5.0 v											OUT	IN	IN			$3B_1$ to 3Y			
			121	GND								OUT		IN						$4B_0$ to 4Y			
			122	5.0 v								OUT	IN							$4B_1$ to 4Y			
11	Same tests, terminal conditions and limits as subgroup 10, except T_C = -55°C.																						

NOTE:

1. INPUTS: A = 2.0 V minimum, B = 0.8 V maximum.
   OUTPUTS: Output voltages shall be either:
   (a) H ≥ 2.4 volts minimum and L ≤ 0.4 volt maximum when using a high speed checker double comparator, or
   (b) H ≥ 1.5 volts and L < 1.5 volts when using a high speed checker single comparator.
   (c) Only attributes data is required for subgroups 7 and 8.

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TABLE III. Group A inspection for device type 06.
Terminal conditions (pins not designated may be H $\geq$ 2.0 V, or L $\leq$ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Case E, F Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Meas. terminal	Test limits Min	Max	Unit
				D_3	D_2	D_1	D_0	Y	W	G	GND	C	B	A	D_7	D_6	D_5	D_4	V_{CC}				
1 T_C = 25°C	V_{OH}	3006	1				2.0 V	-0.8mA		0.8 V	GND	0.8 V	0.8 V	0.8 V					4.5 V	Y	2.4		V
	V_{OH}	3006	2						-0.8mA	2.0 V		2.0 V	2.0 V	2.0 V						W	2.4		
	V_{OL}	3007	3					16 mA		2.0 V		2.0 V	2.0 V	2.0 V						Y		0.4	
	V_{OL}	3007	4				2.0 V		16 mA	0.8 V		0.8 V	0.8 V	0.8 V						W		0.4	
	V_{IC}		5				-12 mA													D0		1.5	
			6			-12 mA														D1			
			7		-12 mA															D2			
			8	-12 mA																D3			
			9															-12 mA		D4			
			10														-12 mA			D5			
			11													-12 mA				D6			
			12												-12 mA					D7			
			13							-12 mA										G			
			14											-12 mA						A			
			15										-12 mA							B			
			16									-12 mA								C			
	I_{IL}	3009	17							0.4 V		5.5 V	5.5 V	5.5 V					5.5 V	G	-0.7	-1.6	mA
			18							GND		5.5 V	5.5 V	0.4 V						A			
			19									5.5 V	0.4 V	5.5 V						B			
			20									0.4 V	5.5 V	5.5 V						C			
			21				0.4 V					GND	GND	GND						D0			
			22			0.4 V							GND	5.5 V						D1			
			23		0.4 V								5.5 V	GND						D2			
			24	0.4 V									5.5 V	5.5 V						D3			
			25									5.5 V	GND	GND				0.4 V		D4			
			26										GND	5.5 V			0.4 V			D5			
			27										5.5 V	GND		0.4 V				D6			
			28										5.5 V	5.5 V	0.4 V					D7			
	I_{IH1}	3010	29							2.4 V		GND	GND	GND						G		40	µA
			30							5.5 V		GND	GND	2.4 V						A			
			31									GND	2.4 V	GND						B			
			32									2.4 V	GND	GND						C			
			33				2.4 V					5.5 V	5.5 V	5.5 V						D0			
			34			2.4 V							5.5 V	GND						D1			
			35		2.4 V								GND	5.5 V						D2			
			36	2.4 V									GND	GND						D3			
			37									GND	5.5 V	5.5 V				2.4 V		D4			
			38										5.5 V	GND			2.4 V			D5			
			39										GND	5.5 V		2.4 V				D6			
			40											GND	2.4 V					D7			
	I_{IH2}		41											GND						G		100	
			42											5.5 V						A			
			43										5.5 V	GND						B			
			44									5.5 V	GND	GND						C			
			45				5.5 V						5.5 V	5.5 V						D0			
			46			5.5 V							5.5 V	GND						D1			
			47		5.5 V								GND	5.5 V						D2			
			48	5.5 V									GND	GND						D3			
			49									GND	5.5 V	5.5 V				5.5 V		D4			
			50									GND	5.5 V	GND			5.5 V			D5			

See note at end of device type 06.

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TABLE III. Group A inspection for device type 06 - Continued.

Terminal conditions (pins not designated may be H, 2.0 V ≤ V_{IH} ≤ V_{CC}, 0.8 V ≥ V_{IL} ≥ GND or open)

Subgroup	Symbol	MIL-STD-883 method	Case E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Meas. terminal	Test limits Min	Test limits Max	Unit
			Test No.	D_3	D_2	D_1		D_0	W	G	GND	C	H	A		D_6	D_7	D_4	V_{CC}			
1, 2, 3	I_{IH2}	3010	51							5.5 V	GND			5.5 V		5.5 V	5.5 V		5.0 V	D_6		100	μA
T_C = 25°C	I_{IH2}	3010	52							5.5 V	GND			GND		GND	GND			D_7		100	μA
	I_{OS}	3011	53	GND	GND	GND		GND	GND	GND	GND							GND		W	20	55	mA
	I_{OS}	3011	54	GND	GND	GND		5.5 V	GND	GND	GND							GND		Y	20	55	mA
	I_{CC}	3005	55	GND	GND	GND		5.5 V	GND	GND	GND							GND		V_{CC}		40	mA
2	Same tests, terminal conditions and limits as subgroup 1 except T_C = 125°C and V_{IC} tests are omitted																						
3	Same tests, terminal conditions and limits as subgroup 1 except T_C = 55°C and V_{IC} tests are omitted																						
7, 8, 9, 10, 11	Truth table test		56		H				H	A	GND	H	H	H									
T_C = 25°C, 125°C, 55°C			57		A				H	H		H	A	H									
			58						H			H	H	A									
			59		H				L			L	H	H									
			60		A				L			L	A	H									
			61						L			L	H	A									
			62	H					H			H	A	A									
			63	A					H			H	H	H									
			64						L			L	A	A									
			65						L			L	H	H									
			66						L			L	A	A									
			67						L			L	H	H									
			68						L			L	A	A									
			69						L			L	H	H									
			70						L			L	A	A									
			71						L			L	H	H									
			72						L			L	A	A									
8	Repeat subgroup 7 at T_C = 125°C and T_C = 55°C																						
9	t_{PHL1}	3003 (Fig 6)	73		5.0 V			GND	OUT	GND	GND	GND	GND	IN					5.0 V	A to W		32	ns
25°C			74									IN	IN	GND						H to W			
			75									GND	GND	IN						C to W			
	t_{PLH1}		76		5.0 V							GND	GND	GND						A to W		29	
			77									IN	IN	GND						H to W			
			78									GND	GND	IN						C to W			
	t_{PHL2}		79		5.0 V				OUT			GND	GND	IN						A to Y		40	
			80									IN	IN	GND						H to Y			
			81									GND	GND	IN						C to Y			
	t_{PLH2}		82		5.0 V							GND	GND	IN						A to Y		30	
			83									IN	IN	GND						H to Y			
			84									GND	GND	IN						C to Y			
	t_{PHL3}		85					5.0 V	OUT	IN										D6 to W		28	
	t_{PLH3}		86						OUT											D7 to W		26	
	t_{PHL4}		87																	D2 to W		37	
	t_{PLH4}		88						OUT											D3 to W		35	
	t_{PHL5}		89					IN		GND										D1 to W		20	
			90																				
			91																				
			92																				

See note at end of device type 06.

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TABLE III. Group A inspection for device type 06 - Continued.

Terminal conditions (pins not designated may be $H \geq 2.0$ V, or $L \leq 0.8$ V, or open)

Subgroup	Symbol	MIL-STD-883 method	Case E_1, F / Test No	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Meas. terminal	Test limits Min	Test limits Max	Test limits Unit
				D3	D2	D1	D0	Y	W	G	GND	C	B	A	D7	D6	D5	D4	V_{CC}				
9	t_{PHL5}	3003	93						OUT	GND	GND	5.0 V	GND	GND				IN	5.0 V	D1 to W	3	20	ns
T_C = 25°C		(Fig 6)	94										GND	5.0 V			IN			D5 to W			
			95										5.0 V	GND		IN				D6 to W			
	t_{PLH5}		96										5.0 V	5.0 V	IN					D7 to W			
			97				IN					GND	GND	GND						D0 to W		17	
			98			IN							GND	5.0 V						D1 to W			
			99		IN								5.0 V	GND						D2 to W			
			100	IN									5.0 V	5.0 V						D3 to W			
			101									5.0 V	GND	GND				IN		D4 to W			
			102										GND	5.0 V			IN			D5 to W			
			103										5.0 V	GND		IN				D6 to W			
			104										5.0 V	5.0 V	IN					D7 to W			
	t_{PHL6}		105				IN	OUT				GND	GND	GND						D0 to Y	6	29	
			106			IN							GND	5.0 V						D1 to Y			
			107		IN								5.0 V	GND						D2 to Y			
			108	IN									5.0 V	5.0 V						D3 to Y			
			109									5.0 V	GND	GND				IN		D4 to Y			
			110										GND	5.0 V			IN			D5 to Y			
			111										5.0 V	GND		IN				D6 to Y			
			112										5.0 V	5.0 V	IN					D7 to Y			
	t_{PLH6}		113				IN					GND	GND	GND						D0 to Y		33	
			114			IN							GND	5.0 V						D1 to Y			
			115		IN								5.0 V	GND						D2 to Y			
			116	IN									5.0 V	5.0 V						D3 to Y			
			117									5.0 V	GND	GND				IN		D4 to Y			
			118										GND	5.0 V			IN			D5 to Y			
			119										5.0 V	GND		IN				D6 to Y			
			120										5.0 V	5.0 V	IN					D7 to Y			
10	t_{PHL1}		121			5.0 V	GND		OUT			GND	GND	IN						A to W		48	
T_C = 125°C			122		5.0 V							GND	IN	GND						B to W			
			123									IN	GND	GND				5.0 V		C to W			
	t_{PLH1}		124			5.0 V						GND	GND	IN						A to W		43	
			125		5.0 V							GND	IN	GND						B to W			
			126									IN	GND	GND				5.0 V		C to W			
	t_{PHL2}		127			5.0 V		OUT				GND	GND	IN						A to Y	8	60	
			128		5.0 V							GND	IN	GND						B to Y			
			129									IN	GND	GND				5.0 V		C to Y			
	t_{PLH2}		130			5.0 V						GND	GND	IN						A to Y		58	
			131		5.0 V							GND	IN	GND						B to Y			
			132									IN	GND					5.0 V		C to Y			
	t_{PHL3}		133				5.0 V		OUT	IN		GND								G to W	6	38	
	t_{PLH3}		134						OUT											G to W	6	35	
	t_{PHL4}		135					OUT												G to Y	8	52	
	t_{PLH4}		136					OUT												G to Y	8	52	
	t_{PHL5}		137				IN		OUT	GND										D0 to W	3	32	
			138			IN								5.0 V						D1 to W			
			139		IN								5.0 V	GND						D2 to W			
			140	IN									5.0 V	5.0 V						D3 to W			

See end of table of device type 06.

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TABLE III. Group A inspection for device type 06 - Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Case E_L, F Test No.	1 D_3	2 D_2	3 D_1	4 D_0	5 Y	6 W	7 G	8 GND	9 C	10 B	11 A	12 D_7	13 D_6	14 D_5	15 D_4	16 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
10 T_C = 125°C	t_{PHL5}	3003 (Fig. 6)	141						OUT	GND	GND	5.0 V	GND	GND				IN	5.0 V	D_4 to W	3	32	ns
			142										GND	5.0 V			IN			D_5 to W			
			143										5.0 V	GND		IN				D_6 to W			
			144										5.0 V	5.0 V	IN					D_7 to W			
	t_{PLH5}		145				IN					GND	GND	GND						D_0 to W		26	
			146			IN							GND	5.0 V						D_1 to W			
			147		IN								5.0 V	GND						D_2 to W			
			148	IN									5.0 V	5.0 V						D_3 to W			
			149									5.0 V	GND	GND				IN		D_4 to W			
			150										GND	5.0 V			IN			D_5 to W			
			151										5.0 V	GND		IN				D_6 to W			
			152										5.0 V	5.0 V	IN					D_7 to W			
	t_{PHL6}		153				IN	OUT				GND	GND	GND						D_0 to Y	6	44	
			154			IN							GND	5.0 V						D_1 to Y			
			155		IN								5.0 V	GND						D_2 to Y			
			156	IN									5.0 V	5.0 V						D_3 to Y			
			157									5.0 V	GND	GND				IN		D_4 to Y			
			158										GND	5.0 V			IN			D_5 to Y			
			159										5.0 V	GND		IN				D_6 to Y			
			160										5.0 V	5.0 V	IN					D_7 to Y			
	t_{PLH6}		161				IN					GND	GND	GND						D_0 to Y		36	
			162			IN							GND	5.0 V						D_1 to Y			
			163		IN								5.0 V	GND						D_2 to Y			
			164	IN									5.0 V	5.0 V						D_3 to Y			
			165									5.0 V	GND	GND				IN		D_4 to Y			
			166										GND	5.0 V			IN			D_5 to Y			
			167										5.0 V	GND		IN				D_6 to Y			
			168										5.0 V	5.0 V	IN					D_7 to Y			
11	Same tests, terminal conditions and limits as subgroup 10, except T_C = -55°C.																						

NOTE 1. INPUTS: A = 2.0 V minimum, B = 0.8 V maximum.

OUTPUTS: Output voltages shall be either:

(a) H ≥ 2.4 volts minimum and L ≤ 0.4 volt maximum when using a high speed checker double comparator, or
(b) H ≥ 1.5 volts and L ≤ 1.5 volts when using a high speed checker single comparator.
(c) Only attributes data is required for subgroups 7 and 8.

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b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.
c. Requirement for certificate of compliance, if applicable.
d. Requirements for notification of change of product or process to the contracting activity in addition to notification to qualifying activity, if applicable.
e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action and reporting of results, if applicable.
f. Requirements for product assurance options.
g. Requirements for carriers, special lead lengths or lead forming, if applicable. These requirements shall not affect the part number.
h. Requirements for JAN marking.

6.4 Abbreviations, symbols and definitions. The abbreviations, symbols, and defintions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND	Electrical ground (common terminal)
I_{IN}	Current flowing into an input terminal
V_{IC}	Input clamp voltage
V_{IN}	Voltage level at an input terminal

6.5 Logistic support. Lead materials and finishes (see 3.3), are interchangeable. Unless otherwise specified, microcircuits acquired to Government logistic support will be acquired to device class B (see 1.2.2), and lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.6 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a wavier of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	54150
02	9312
03	54153
04	9309
05	9322, 54157
06	54151

6.7 Manufacturers designation. Manufacturer circuits included in this specification are designated as shown in table IV herein.

TABLE IV. Substitutability and manufacturers designator.

Device Types	Motorola	Signetics	Fairchild	Texas Instruments	National	Advanced Micro Device
	A	B	C	D	E	F
01	X	X				
02	X	X				X
03	X	X	X		X	
04	X	X	X			X
05	X	X	X	X	X	X
06	X	X	X	X	X	

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6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - SH, OS
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Agent:
DLA - ES

Preparing activity:
Air Force - 17

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4.1.1 Specifications, standards, and handbooks. The following specifications, standards, and

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(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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